Business Architecture

Table of contents

Business Architecture **3**
- Purpose of this chapter . . . 3
- Position in the framework . . . 3
- How NSIs use this chapter . . . 4
- Relationship with the statistical standards . . . 5
- Business architecture concepts in the SAF . . . 5
- How the business architecture is built up . . . 6
  - Functions and processes . . . 7
  - Processes and the data value chain . . . 7
  - Generic and model-specific content . . . 8
- Structure of this chapter . . . 8
- Business Actors, Business Roles and Business Collaborations . . . 8
  - The concepts . . . 8
  - How the SAF applies actors, roles and collaborations . . . 9
  - SAF business actors . . . 9
  - SAF business roles . . . 13
  - SAF business collaborations . . . 23
  - Role names in the value streams . . . 27
- Business Processes . . . 28
  - The concept . . . 28
  - How the SAF applies business processes . . . 28
  - A. Statistical production processes . . . 30
  - B. Overarching processes . . . 41
  - C. Data management and lifecycle processes . . . 43
  - D. Microdata services processes . . . 45
  - E. Data space processes . . . 48
- Business Functions . . . 50
  - The concept . . . 50
  - How the SAF applies business functions . . . 50
  - FUN-1 Strategy and Leadership . . . 51
  - FUN-2 Capability Development . . . 53
  - FUN-3 Statistical Production . . . 55
  - FUN-4 Methodology and Research . . . 57
  - FUN-5 Quality Management . . . 58
  - FUN-6 Data Supplier Management . . . 59
  - FUN-7 Data Management . . . 60
  - FUN-8 Microdata Services . . . 62
  - FUN-9 Data Space Participation . . . 64
  - FUN-10 Business Support . . . 65
  - Process and function mapping . . . 68
  - Mapping to value stream stages . . . 71
- Business Services and Business Interfaces . . . 74
- Business Events . . . 74
- Business Objects, Contracts, Representations and Products . . . 74

Reference Models for the Statistical Data Value Chain . . . 74
Comparison . . . 75
Selection criteria . . . 75
Combining the models . . . 76

Steady-State Architecture **76**
Overview . . . 76
Business domains . . . 76
The four process types . . . 77
Design rules of the steady-state model . . . 77
Steady states at a glance . . . 78
State lifecycle . . . 78
How each state is described . . . 78
Boundary Position #0: Source Data . . . 79
State #1: Raw Data . . . 79
1. Definition and value added . . . 79
2. Producing and consuming domains . . . 79
3. Quality gate . . . 80
4. Data and metadata characteristics . . . 80
5. Security, privacy, and access . . . 81
6. Ownership and governance . . . 81
7. Versioning, retention, and lineage . . . 81
8. Reuse and handover . . . 81
9. Traceability to the framework . . . 82
10. Examples and anti-patterns . . . 82
State #2: Standardized Data . . . 82
1. Definition and value added . . . 82
2. Producing and consuming domains . . . 83
3. Quality gate . . . 83
4. Data and metadata characteristics . . . 84
5. Security, privacy, and access . . . 84
6. Ownership and governance . . . 85
7. Versioning, retention, and lineage . . . 85
8. Reuse and handover . . . 85
9. Traceability to the framework . . . 86
10. Examples and anti-patterns . . . 86
State #3: Processed Data . . . 87
1. Definition and value added . . . 87
2. Producing and consuming domains . . . 87
3. Quality gate . . . 87
4. Data and metadata characteristics . . . 88
5. Security, privacy, and access . . . 89
6. Ownership and governance . . . 89
7. Versioning, retention, and lineage . . . 90
8. Reuse and handover . . . 90
9. Traceability to the framework . . . 90
10. Examples and anti-patterns . . . 91
State #4: Statistics . . . 91
1. Definition and value added . . . 91
2. Producing and consuming domains . . . 92
3. Quality gate . . . 92
4. Data and metadata characteristics . . . 93
5. Security, privacy, and access . . . 94
6. Ownership and governance . . . 94
7. Versioning, retention, and lineage . . . 95
8. Reuse and handover . . . 95
9. Traceability to the framework . . . 95
10. Examples and anti-patterns . . . 96

State #5: Released Data . . . 96
1. Definition and value added . . . 96
2. Producing and consuming domains . . . 97
3. Quality gate . . . 97
4. Data and metadata characteristics . . . 98
5. Security, privacy, and access . . . 98
6. Ownership and governance . . . 98
7. Versioning, retention, and lineage . . . 99
8. Reuse and handover . . . 99
9. Traceability to the framework . . . 99
10. Examples and anti-patterns . . . 100
Boundary Position #6: Consumed Data . . . 100
The Statistical Data Value Chain . . . 101
Steady States of Data . . . 101
Applying the steady states pragmatically . . . 103
The Four-Process Governance Pattern . . . 105
Principles to Apply . . . 105
VS-01.0 — Data Acquisition and Ingestion . . . 106
VS-02.0 — Data Standardization and Integration . . . 107
VS-03.0 — Data Editing and Enrichment . . . 108
VS-04.0 — Estimation, Analysis, and Statistical Modeling . . . 109
VS-05.0 — Statistical Product Composition . . . 110
VS-06.0 — Dissemination and Service Delivery . . . 111
Supporting Value Streams (VS-07 to VS-10) . . . 112
VS-07.0 — Governance and Compliance . . . 113
VS-08.0 — Innovation and Methodological Development . . . 114
VS-09.0 — Infrastructure and Platform Enablement . . . 115
VS-10.0 — Customer and Stakeholder Engagement . . . 116
The Value Chain at a Glance . . . 116

Business Architecture

Purpose of this chapter

The business architecture describes how a statistical organisation is organised to turn data into official statistics. It shows who is involved, which processes and functions they perform, which services they offer, what triggers their work, and which information they create and exchange.

The Strategy layer describes *what* the NSI must be able to do and *where* it creates value. The business architecture describes *how* that work is organised. It is the layer where capabilities and value streams become concrete: in roles that people take on, processes they carry out, and information they hand over to each other.

For official statistics, one question deserves special attention: where does responsibility for data pass from one part of the organisation to another? A statistical value chain is long. It involves many teams and often confidential data. An untraceable change or uncontrolled reuse at any point can undermine every statistic built on that data. The SAF business architecture therefore pays explicit attention to the points where data is stored, quality-assured and handed over. These handover points are the backbone of the reference models at the end of this chapter.

The business architecture is independent of technology. It does not describe applications or platforms. It does describe the business needs that applications and platforms must support, and it is the starting point for the data, application and technology architecture.

Position in the framework

The business architecture is the first of the core layers in TOGAF and ArchiMate (see *Position in the framework* in the Motivation chapter). In the TOGAF Architecture Development Method, it is developed

in Phase B, Business Architecture. It receives direction from the Strategy layer and gives direction to the layers below it.

PLACEHOLDER: Position of the Business Architecture between the Strategy layer and the data, application and technology layers

Direction	Relationship	Example in the SAF
From Strategy	Business processes realise value stream stages	The execution processes of the Standardization domain realise the stages of VS-02
From Strategy	Business functions and processes realise capabilities	The function *Classify Data* realises Data standardisation (1.3.1)
From Motivation	Principles and requirements shape the business design	SAF-P07 leads to loose coupling of domains through steady states
To Data Architecture	Business objects are realised by data objects	A registered *Raw Data* version is realised by a dataset in a data store
To Application Architecture	Business processes are served by application services	Data validation at intake is served by a validation service
To Technology Architecture	Business requirements drive platform requirements	Retention and versioning per steady state drive storage requirements

In TOGAF, Phase B produces a baseline and a target business architecture, and the gap between them. The SAF provides a reference for the target. An NSI describes its own baseline and uses the SAF to define where it wants to go.

How NSIs use this chapter

The SAF business architecture contains two kinds of content.

Reference content. Most NSIs perform the same statistical work and follow a comparable production process. The SAF therefore offers reference content for business processes, business functions, business services and business roles. NSIs use this content in the same three ways as the other SAF elements: adopt, adapt or extend.

Concepts with examples. Business events, business objects, contracts, representations and products are specific to each NSI. They depend on its products, its legal basis and its users. For these elements, the SAF explains the concept, how it is used, and gives examples. The toolkit provides templates.

The data value chain needs a separate choice. NSIs organise their chain in different ways, depending on their size, legal context and platforms. The SAF therefore offers more than one reference model for the chain: the Steady-State Architecture and the Medallion Architecture. An NSI chooses the model that fits, or combines them. This choice is a course of action in the sense of the Strategy chapter.

The business architecture is also the place where the NSI's own organisation meets the framework. An NSI maps its departments, teams and job profiles onto the SAF actors, roles and functions. The mapping shows which parts of the organisation carry which responsibilities, where they overlap, and where they are missing.

PLACEHOLDER: Business architecture templates and mapping instruments to be added to the toolkit

Relationship with the statistical standards

The SAF business architecture builds on two models that the statistical community already uses.

GSBPM, the Generic Statistical Business Process Model, describes statistical production in phases and sub-processes, from *Specify Needs* to *Evaluate*. The SAF process model is based on GSBPM.

GAMSO, the Generic Activity Model for Statistical Organisations, describes the activities of a statistical organisation as a whole: strategy and leadership, capability development, production, and the activities that support them. GAMSO places GSBPM in a wider context: GSBPM describes the production activities of GAMSO in detail. The SAF uses GAMSO as the frame of reference for its business functions. The functions are described in SAF terms, so that NSIs can apply them together with their own national function models. An NSI that already works with GAMSO will recognise the main areas.

GSIM, the Generic Statistical Information Model, describes the information objects used in statistical production. It is the reference for the SAF business objects.

Business architecture concepts in the SAF

The SAF uses the business layer elements of ArchiMate, and adds concepts that are specific to the statistical data value chain.

ArchiMate business elements used in the SAF

Concept	ArchiMate element	Question it answers	Example	SAF content
Business actor	Business Actor	Who is involved: a person, team or organisation?	Data provider, NSI, research institute	Reference
Business role	Business Role	Which responsibility does an actor take on?	Data owner, data steward, methodologist	Reference
Business collaboration	Business Collaboration	Which actors or roles work together?	A data sharing partnership between an NSI and a register holder	Reference
Business process	Business Process	Which sequence of activities produces a defined result?	Collect, Process, Disseminate	Reference
Business function	Business Function	What does the organisation do continuously, grouped by required knowledge and skills?	Manage Quality	Reference
Business service	Business Service	Which behaviour does the organisation offer to others?	Microdata access for researchers	Reference
Business interface	Business Interface	Through which channel is a service offered?	A web portal, a secure data transfer channel, a helpdesk	Reference
Business event	Business Event	What triggers a process, or marks its result?	*Raw Data Registered*; start of a reference period	Concept and examples
Business object	Business Object	Which information does the business use and produce?	Raw Data, Statistical Product, Classification	Concept and examples

Concept	ArchiMate element	Question it answers	Example	SAF content
Contract	Contract	Which formal agreement governs a service or exchange?	A data provision agreement with a register holder	Concept and examples
Representation	Representation	In which perceptible form is information made available?	A press release, a statistical table, an API response	Concept and examples
Product	Product	Which coherent set of services and information is offered to users?	A statistical product with its publication and data services	Concept and examples

SAF-specific concepts for the data value chain

Concept	Definition
Business domain	A coherent part of the statistical process that adds one specific kind of statistical value to data and is governed as a unit. In ArchiMate, a business domain is modelled as a grouping.
Data state	A defined condition of data, characterised by the processing it has undergone and the quality criteria it meets.
Steady state	A data state that is stored, versioned, quality-assured and registered, so that it can be handed over and reused.
Quality gate	The set of criteria that data must meet before it may be registered in a state.
Quality version	An immutable, identified version of a dataset that has passed a quality gate.
Handover interface	The agreement between a producing and a consuming domain: the artefact, its required metadata, the triggering event and the transfer of responsibility.
Reference model	A generic arrangement of domains, states and handover points that an NSI can adopt and adapt, such as the Steady-State or Medallion Architecture.

How the business architecture is built up

ArchiMate describes each layer with three aspects: *who* acts (active structure), *what* happens (behaviour) and *what* is acted upon (passive structure). The SAF business architecture follows these aspects, and then brings them together in two integrating views.

PLACEHOLDER: Structure of the SAF business architecture: three ArchiMate aspects and two integrating views

Part	Elements	Question
Active structure	Actors, roles, collaborations, interfaces	*Who* does the work, and through which channels?

Part	Elements	Question
Behaviour	Processes, functions, services, events	*What* happens, in which order, and what triggers it?
Passive structure	Business objects, contracts, representations, products	*Which* information is used, agreed and delivered?
Integrating view 1	Process and function architecture and value chain	*How* do processes, functions and value streams fit together?
Integrating view 2	Steady-State and Medallion Architectures	*Where* is data handed over, and with which quality?

Functions and processes

Functions and processes are easily confused, but they are different.

A **business function** groups behaviour by the knowledge, skills and resources it needs. Functions are stable. They have no start or end, and they do not depend on the order of work. *Manage Quality* is a function: it exists as long as the NSI exists.

A **business process** is a sequence of activities that leads to a defined result. A process has a trigger, a start and an end. *Collect* is a process: it starts when a collection period opens and ends when the data is stored and checked.

The two cross each other. One process uses several functions, and one function contributes to several processes. In ArchiMate, functions and processes are connected through realisation and assignment relationships. Statistical production is the clearest example: the production function is realised by the production process, which the SAF describes in detail.

Processes and the data value chain

The process view and the data value chain look at the same production from two angles. The process follows the life cycle of one statistical product, from need to evaluation. The data value chain follows the data, from source to release, across products.

In the Steady-State Architecture, each business domain uses four process types. They connect the two angles:

Process type in a business domain	Provides	Corresponding GSBPM phases
Design	Metadata: designs, rules, specifications	2 Design
Implementation	The execution process: instruments, components, configured workflows	3 Build
Execution	Statistical data, registered in a steady state	4 Collect, 5 Process, 6 Analyse, 7 Disseminate
Management	Quality reports and improvements	8 Evaluate, and the overarching quality and metadata management

The GSBPM phases 4 to 7 run across several business domains. *Process* (phase 5), for example, spans standardization, editing and part of estimation. The steady states mark where, within that flow, data is handed over with a defined quality.

Generic and model-specific content

Actors, roles, processes, functions, services, events and objects are generic. They apply to every NSI, whatever reference model it chooses for its data value chain.

The data value chain is model-specific. The business domains, steady states and quality gates differ between the Steady-State and the Medallion Architecture. That is why the SAF describes them per reference model, at the end of this chapter.

Structure of this chapter

Section	Content	SAF content
Business Actors, Business Roles and Business Collaborations	Who is involved in statistical production, and with which responsibilities	Reference
Business Processes	The SAF process model, based on GSBPM	Reference
Business Functions	The SAF function model	Reference
Business Services and Business Interfaces	The services an NSI offers, and the channels through which it offers them	Reference
Business Events	Triggering events: what they are, how they are used, examples	Concept and examples
Business Objects, Contracts, Representations and Products	Information objects, agreements, forms of output and products: what they are, how they are used, examples	Concept and examples
Process and Function Architecture and Value Chain	How processes, functions and value streams fit together, per value stream	Reference
Steady-State and Medallion Architectures	Reference models for the data value chain: comparison, selection criteria, combination, and each model in detail	Reference

Readers who want to understand the statistical data value chain can go straight to the last section. Readers who want to map their own organisation onto the SAF start with actors and roles, and with the function model.

Business Actors, Business Roles and Business Collaborations

The concepts

A **business actor** is a person, team or organisation that can perform behaviour. An actor is a real entity. It exists whether or not it takes part in a specific process. Examples are a ministry that supplies a register, a research institute, or the NSI itself. In the SAF, actors are described as types, not as named organisations.

A **business role** is a responsibility for performing specific behaviour. A role says what someone is accountable for, not who that someone is. One actor can take on several roles, and one role can be filled by several actors. A methodologist in a small NSI may also act as statistical designer. A large NSI may have a whole team in the role of data steward.

A **business collaboration** is a group of actors or roles that work together for a shared purpose. A collaboration performs behaviour that none of its members can perform alone. It can be permanent, such

as a data governance board, or temporary, such as a project team. Collaborations can also cross the boundaries of the NSI. A data sharing partnership between an NSI and a register holder is an example.

Actors, roles and collaborations are different from stakeholders (see *Stakeholders* in the Motivation chapter). A stakeholder has an interest in the architecture. An actor or role performs work in it. The same organisation can be both. A data provider is a stakeholder with concerns about burden, and an actor that delivers data in VS-01.

How the SAF applies actors, roles and collaborations

Why the SAF describes actors, roles and collaborations

Roles make responsibilities explicit. A steady state needs a data owner. A quality gate needs someone who decides. A release needs someone who approves. Without defined roles, these decisions are taken by whoever happens to be involved. That makes accountability unclear and audits difficult. Defined roles also make NSIs comparable. When two NSIs both use the role *data steward*, they can compare practices and share solutions.

How actors, roles and collaborations fit into an NSI's own architecture

The SAF offers a reference set of actor types, roles and collaborations. An NSI maps its own organisation onto this set. It links its departments and teams to actor types, and its job profiles to roles. One job profile often covers several SAF roles, especially in small NSIs. The mapping shows which roles are filled, which are combined, and which are missing. An NSI can then assign roles to processes, for example with a RACI matrix (responsible, accountable, consulted, informed).

The SAF roles are deliberately generic. The value streams use more specific role names, such as *Estimation Methodologist* or *Imputation Specialist*. These are specialisations of the reference roles. The mapping table at the end of this section shows how they relate.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (Motivation)	A stakeholder can also be an actor	A data provider is a stakeholder and an actor in VS-01
Business processes	Roles are assigned to processes	The data steward is assigned to *Register Raw Data*
Business functions	Roles are assigned to functions	The quality manager is assigned to *Manage Quality*
Business services and interfaces	Actors and roles offer or use services through interfaces	A researcher uses the microdata access service through a secure portal
Steady states	Each steady state has an owner, a steward and a custodian	The data owner of State #3 decides on access to processed microdata
Capabilities (Strategy)	Roles are part of the resources that realise capabilities	Methodologists are a key resource for Methodology (2.2.3)

PLACEHOLDER: ArchiMate view of SAF actors, roles and collaborations

SAF business actors

The SAF distinguishes nine actor types. They are grouped by their position relative to the NSI.

ID	Actor	Position	Description
ACT-01	National Statistical Institute	Internal	The NSI is the organisation that produces official statistics under a legal mandate. It is the central actor in the SAF. The NSI owns the statistical processes and is accountable for the quality and confidentiality of its output. It enters into agreements with data providers, users and partners. In the architecture, the NSI as a whole is the actor that offers business services to society.
ACT-02	Organisational unit	Internal	An organisational unit is a department, division or team within the NSI. Units are the actors that actually carry out the work. Each unit fills one or more roles, such as data collection, methodology or dissemination. The SAF does not prescribe an organisation structure. An NSI maps its own units onto the SAF roles, so that it can see which unit is responsible for which part of the chain.
ACT-03	Data provider	External, supplies data	A data provider is an organisation that supplies data it holds to the NSI. Examples are government agencies with administrative registers, businesses with transaction data, and platform operators. The data is collected for another purpose and reused for statistics. Delivery usually takes place under a legal obligation or a data provision agreement. The NSI cannot control the quality at the source, but it can influence it through agreements and feedback.

ID	Actor	Position	Description
ACT-04	Respondent	External, supplies data	A respondent is a person, household or business that provides data directly to the NSI, usually through a survey. Respondents answer questionnaires on the web, by telephone, face to face or on paper. Their burden is a constant concern for the NSI. Respondents are protected by statistical confidentiality: the NSI may not disclose their individual data. The NSI also has a duty to explain why it needs the data and how it uses it.
ACT-05	Data user	External, uses output	A data user is a person or organisation that uses the output of the NSI. Users include policy makers, researchers, journalists, businesses and members of the public. Their needs differ widely. A minister needs a clear figure, a researcher needs microdata under strict conditions, and a developer needs a reliable API. The NSI groups users into segments, so that it can design services and channels for each segment.

ID	Actor	Position	Description
ACT-06	Other producer of official statistics	Statistical system	Many countries have more than one producer of official statistics. Examples are central banks, ministries with statistical units and regional statistical offices. The NSI often coordinates this national statistical system. Producers exchange data, align methods and share infrastructure. In the architecture, they are partners in collaborations and sometimes providers or users of each other's data.
ACT-07	International statistical organisation	Statistical system	International statistical organisations set standards, coordinate statistics across countries and collect national data. Examples are UNECE, Eurostat, the OECD, the IMF and the UN Statistics Division. They require NSIs to deliver data in agreed formats and on agreed dates. They also develop the shared models that the SAF builds on, such as GSBPM and GSIM. For the NSI, they are users, standard setters and partners in collaboration.

ID	Actor	Position	Description
ACT-08	Supervisory and legal authority	External, oversees	A supervisory or legal authority oversees whether the NSI acts within the law. Examples are the data protection authority, a national statistics council, the national audit office and the courts. Some authorities grant the legal basis for data collection. Others assess compliance after the fact. The NSI must be able to show these authorities how it handles data, which is one reason why traceability is a core concern of the SAF.
ACT-09	Supplier and service provider	External, supports	A supplier delivers goods or services that the NSI needs but does not produce itself. Examples are software vendors, cloud and hosting providers, field work agencies and trusted third parties for pseudonymisation. Suppliers work under contracts that set requirements for security, confidentiality and continuity. When a supplier processes statistical data, the NSI remains accountable for it. The choice of suppliers is guided by SAF-P01 and by the NSI's sourcing strategy.

SAF business roles

The SAF reference set contains 24 roles in seven groups. The column *Process type* shows in which of the four process types of a business domain (design, implementation, execution, management) the role mainly works.

Governance and accountability

ID	Role	Process type	Description
ROL-01	Process owner	Management	The process owner is accountable for the end-to-end performance of a business domain or value stream. This role sets the objectives of the process and decides on changes to it. The process owner makes sure the process has the people, methods and systems it needs. The role receives the quality reports of the management process and acts on them. In the steady-state model, each business domain has one process owner.
ROL-02	Data owner	Management	The data owner is accountable for a dataset or a steady state. This role decides who may access the data, for which purpose and under which conditions. The data owner also approves the retention period and the classification of the data. The role is usually held by a manager in the domain that produces the data. Data ownership is a business role, not an IT role: the data owner answers for the content, not for the storage.
ROL-03	Data steward	Execution	The data steward takes care of data on behalf of the data owner. This role registers datasets in the catalogue and maintains their metadata. The data steward checks that data meets the quality gate before it is registered. The role also handles questions from users and resolves exceptions. Data stewards are the daily point of contact for anyone who wants to find or use a dataset.

ID	Role	Process type	Description
ROL-04	Data custodian	Execution	The data custodian is responsible for the safe storage and technical management of data. This role runs storage, backup and access control according to the decisions of the data owner. The custodian carries out retention and disposal. The role does not decide on the use of the data, but makes sure that decisions are applied correctly. In many NSIs, the custodian role is held by an IT or platform team.
ROL-05	Chief data officer	Management	The chief data officer is accountable for data governance across the whole NSI. This role sets the data policy and the governance framework. It assigns the data owner, steward and custodian roles to people and units. The chief data officer chairs or supports the data governance board. The role is the escalation point for conflicts about data that cross domain boundaries.

Design

ID	Role	Process type	Description
ROL-06	Statistical designer	Design	The statistical designer translates information needs into the design of a statistical product. This role defines the outputs, the variables, the population and the required data. The designer specifies the processes that produce the product, in close cooperation with the methodologist. The role is the link between users and production. In some NSIs this role is called product manager.
ROL-07	Methodologist	Design	The methodologist designs and assesses the statistical methods. This role covers sampling, collection methods, editing and imputation, estimation and quality measurement. The methodologist makes sure methods are sound, documented and reproducible. The role also evaluates new methods before they enter production. Specialisations exist for each step of the chain, such as estimation or editing.
ROL-08	Metadata and data architect	Design	The metadata and data architect designs the structure of data and metadata. This role defines data models, metadata schemas and the minimum metadata set of each steady state. The architect makes sure that data from different sources can be combined and compared. The role works with standards such as GSIM and SDMX. It is the bridge between the business architecture and the data architecture.

ID	Role	Process type	Description
ROL-09	Classification steward	Design	The classification steward manages the classifications and code lists that the NSI uses. This role maintains national versions of international classifications, such as NACE or ISCO. The steward manages correspondence tables between versions. The role makes sure that all domains use the same version at the same time. Changes to a classification are announced and documented before they take effect.

Implementation

ID	Role	Process type	Description
ROL-10	Data engineer	Implementation	The data engineer builds the technical processes that move and transform data. This role develops ingestion pipelines, collection channels and processing workflows. The data engineer implements the designs of the statistical designer and the methodologist. The role also builds the automated checks that support the quality gates. The data engineer works closely with the platform engineer, who provides the environment.

ID	Role	Process type	Description
ROL-11	Platform engineer	Implementation	The platform engineer provides and runs the environments in which the processes operate. This role configures platforms, storage, connectors and access controls. The platform engineer keeps environments secure, available and up to date. The role supports all value streams but is anchored in VS-09. In many NSIs, the role is also called system configurator or DevOps engineer.
ROL-12	Test manager	Implementation	The test manager plans and coordinates the testing of new or changed processes. This role covers technical tests, integration tests and field tests. The test manager checks that a process produces the expected result before it goes into production. The role reports test results to the process owner. Testing is required before any change to a process that produces a steady state.

Execution

ID	Role	Process type	Description
ROL-13	Data collection manager	Execution	The data collection manager runs data collection campaigns. This role plans the collection period, coordinates field staff and monitors response. The manager follows up on non-response and late deliveries. The role also handles contacts with respondents and data providers during collection. The data collection manager reports on collection progress and quality to the process owner.
ROL-14	Data quality analyst	Execution	The data quality analyst checks data during processing. This role runs validation checks, assesses results against the quality gate and logs exceptions. The analyst investigates errors and outliers and proposes corrections. The role documents quality indicators for each version of a dataset. Data quality analysts are active at every handover point in the chain.
ROL-15	Subject-matter statistician	Execution	The subject-matter statistician has in-depth knowledge of a statistical domain, such as labour market or prices. This role interprets results, checks plausibility and explains developments. The statistician carries out or supervises editing, estimation and analysis in the domain. The role writes or reviews the texts that accompany a release. Subject-matter statisticians are the experts that users and journalists turn to.

ID	Role	Process type	Description
ROL-16	Disclosure control specialist	Design, Execution	The disclosure control specialist protects the confidentiality of respondents in the output. This role designs and applies disclosure control methods, such as suppression, rounding and noise. The specialist assesses the disclosure risk of tables, microdata and services. The role works at State #4 and at the gate to State #5. No output is released without a disclosure check that this role has approved.
ROL-17	Release manager	Execution	The release manager controls the publication of statistical output. This role makes sure that releases follow the release calendar and the embargo rules. The release manager checks that the output has been approved and has passed disclosure control. The role coordinates the release across channels, such as the website, the API and press releases. Equal access for all users at the moment of release is a key responsibility.

Assurance

ID	Role	Process type	Description
ROL-18	Quality manager	Management	The quality manager is responsible for the quality framework of the NSI. This role defines quality indicators and monitors them across the chain. The quality manager produces quality reports and leads improvement actions. The role supports process owners in the management process of their domains. It also prepares the NSI for quality reviews, such as peer reviews against the European Statistics Code of Practice.
ROL-19	Compliance officer	Management	The compliance officer makes sure that the NSI complies with laws and regulations. This role covers statistical law, data protection law and other applicable rules. The officer checks new processes and methods against compliance criteria before they go into production. The role keeps the legal register up to date. In many NSIs the data protection officer is a specific form of this role.
ROL-20	Security officer	Management	The security officer is responsible for information security and continuity. This role defines security measures and monitors their application. The officer manages security incidents and leads recovery when something goes wrong. The role assesses the security of new solutions and suppliers. It works closely with the platform engineer and the compliance officer.

ID	Role	Process type	Description
ROL-21	Auditor	Management	The auditor checks independently whether processes and controls work as intended. This role can be internal, such as an internal audit department, or external, such as the national audit office or a certification body. The auditor reviews evidence, such as logs, quality reports and lineage records. The role reports findings to management and to supervisory authorities. Its work depends on the traceability that the SAF builds into the chain.

Users and partners

ID	Role	Process type	Description
ROL-22	Dissemination and service manager	Design, Execution	The dissemination and service manager designs and manages the services through which users receive output. This role covers publications, open data, APIs and microdata access. The manager defines channels and formats for each user segment. The role manages the portfolio of services and their service levels. It works with the release manager on the timing and with the relationship manager on user needs.

ID	Role	Process type	Description
ROL-23	Relationship manager	Execution	The relationship manager maintains the relationships with key external actors. These include data providers, major users, partners and international organisations. The manager collects needs, feedback and complaints, and passes them to the right roles. The role negotiates and maintains agreements, such as data provision agreements. Relationship managers make sure that the NSI speaks with one voice to each partner.

Support

ID	Role	Process type	Description
ROL-24	Enterprise architect	Design, Management	The enterprise architect develops and maintains the enterprise architecture of the NSI. This role translates strategy into architecture and uses the SAF as its reference. The architect advises on investments and reviews projects against the architecture principles. The role supports the architecture board and keeps the architecture repository up to date. It is the role that realises the capability *Architecture management*.

SAF business collaborations

ID	Collaboration	Members	Type	Description
COL-01	Data provision partnership	NSI, data provider, relationship manager, data owner	External, permanent	A data provision partnership is the long-term cooperation between the NSI and a data provider. It goes beyond a single delivery. The partners agree on content, format, timing and quality of the data. They also agree on how to handle changes at the source, such as a new system or a new definition. A good partnership reduces the burden for the provider and improves the quality for the NSI.
COL-02	National statistical system	NSI, other producers of official statistics	External, permanent	The national statistical system is the cooperation between all producers of official statistics in one country. The NSI usually coordinates it. The members align methods, classifications and release practices, so that national statistics are consistent. They exchange data and sometimes share infrastructure. The cooperation is often based on the national statistics law.

ID	Collaboration	Members	Type	Description
COL-03	International statistical cooperation	NSI, international statistical organisations, other NSIs	External, permanent	International statistical cooperation is the collaboration between NSIs and international organisations. It covers the development of standards, the exchange of data and the joint development of tools. The SAF itself is a product of this cooperation. Members take part in working groups, such as those of UNECE and the European Statistical System. The cooperation helps NSIs avoid building the same solution many times.
COL-04	Research access partnership	NSI, research institutes, data owner, disclosure control specialist	External, permanent	A research access partnership gives researchers access to microdata under strict conditions. The NSI, research institutes and sometimes funding bodies take part. The partners agree on who may access which data, for which purpose and in which environment. Output is checked for disclosure risk before it leaves the secure environment. The partnership supports science while protecting the confidentiality of respondents.

ID	Collaboration	Members	Type	Description
COL-05	Statistical product team	Statistical designer, methodologist, data engineer, subject-matter statistician	Internal, permanent or temporary	A statistical product team brings together the roles needed to design, build and run a statistical product. It typically includes a statistical designer, a methodologist, a data engineer and a subject-matter statistician. The team can be permanent for a product or temporary for a redesign. Working as one team shortens the path from design to production. It also makes sure that methodological and technical choices are made together.
COL-06	Data governance board	Chief data officer, data owners, data stewards, compliance officer	Internal, permanent	The data governance board steers data management across the NSI. The chief data officer chairs it, and data owners, data stewards and the compliance officer take part. The board sets data policies, resolves conflicts between domains and monitors compliance. It decides on access requests that cross domain boundaries. The board makes sure that the steady-state model is applied consistently.

ID	Collaboration	Members	Type	Description
COL-07	Architecture board	Enterprise architect, process owners, CIO or CTO	Internal, permanent	The architecture board steers the enterprise architecture of the NSI. The enterprise architect, process owners and the CIO or CTO take part. The board reviews projects against the architecture principles and the target architecture. It decides on deviations through a documented dispensation process. The board keeps the architecture aligned with the strategy and with the SAF.

Role names in the value streams

The value streams use specific role names that fit their context. The table shows how they relate to the reference roles. A specialised role inherits the responsibilities of its reference role and adds those specific to its value stream.

Role name in the value streams	Reference role
Process Owner; Process Owners (VS-01…VS-10)	ROL-01 Process owner
Data Owner / Steward / Custodian	ROL-02 Data owner; ROL-03 Data steward; ROL-04 Data custodian
Chief Data / Governance Officer	ROL-05 Chief data officer
Statistical Designer; Product Manager; Model Designer; Derivation Designer; Coherence Designer	ROL-06 Statistical designer
Methodologist; Estimation Methodologist; Editing Methodologist; Imputation Specialist; Indicator/Index Methodologist; Head of Methodology / Innovation	ROL-07 Methodologist
Metadata Architect; Data Architect	ROL-08 Metadata and data architect
Classification Steward; Coding Specialist	ROL-09 Classification steward
Data Engineer; Integration Specialist; Integration Engineer; Tool / Platform Engineer	ROL-10 Data engineer
System Configurator; DevOps / Platform Engineer; Platform / Cloud Architect; Service Desk / Operations	ROL-11 Platform engineer
Test Manager	ROL-12 Test manager
Data Collection Manager	ROL-13 Data collection manager
Data Quality Analyst	ROL-14 Data quality analyst

Role name in the value streams	Reference role
Subject-Matter Analyst; Subject-Matter Reviewer; Subject-Matter Author; Subject-Matter Expert; Survey Statistician; Analyst / Modeler; Researcher / Data Scientist	ROL-15 Subject-matter statistician
Disclosure-Control Designer	ROL-16 Disclosure control specialist
Release Manager; Approval Authority	ROL-17 Release manager
Quality Manager	ROL-18 Quality manager
Compliance Officer; Legal Counsel	ROL-19 Compliance officer
Security & Continuity Officer	ROL-20 Security officer
Internal / External Auditor	ROL-21 Auditor
Dissemination Manager; Channel/Service Designer; Open-Data / API Product Owner; Service / Product Portfolio Manager; User-Support Lead; Service Desk / User Support	ROL-22 Dissemination and service manager
Account / Relationship Manager; Engagement / Communications Officer; Communications / Outreach Officer; Head of Communication / Engagement; User Researcher; Impact Analyst	ROL-23 Relationship manager
Chief Technology / Infrastructure Officer	Actor: management of the NSI (ACT-01); member of COL-07
Trainer / Knowledge Manager	Not included in the reference set
Data Provider; Survey Respondent; Data User; Legal Authority; Standards Body; Methodology Authority	Actors ACT-03, ACT-04, ACT-05, ACT-08, ACT-07

PLACEHOLDER: Role mapping and RACI templates to be added to the toolkit

Business Processes

The concept

A **business process** is a sequence of activities that produces a defined result for a customer, a stakeholder or another process. A process has a trigger, a start and an end. It has inputs and outputs, and it uses the knowledge and resources of one or more business functions. Processes describe *how* work flows, step by step. They are the part of the business architecture that people recognise most directly in their daily work.

Processes can be decomposed. A high-level process consists of sub-processes, which can in turn be broken down further. The SAF describes processes at two levels: process phases or process groups, and the sub-processes within them. More detailed levels, such as work instructions, are specific to each NSI and fall outside the SAF.

A process is not the same as a value stream stage. A value stream stage describes the value that is added. A process describes the work that adds it. One stage is usually realised by several processes, and one process can contribute to several stages.

How the SAF applies business processes

Why the SAF describes business processes

Processes are where architecture meets daily operations. A shared process model lets NSIs compare how they work, reuse each other's solutions and discuss improvements in the same terms. It also links the strategic layer to the systems: value streams are realised by processes, and processes are supported by application services. Without a process model, it is hard to show which system supports which part of the work, or where a change in method affects other steps.

How business processes fit into an NSI's own architecture

The core of the SAF process model is GSBPM, which most NSIs already know. The SAF adds three process groups that GSBPM does not describe in detail: data management and lifecycle, microdata services and data spaces. An NSI adopts the process model as a reference, maps its own processes onto it and adds its own lower levels. A national process for, say, the monthly price index can then be traced to the SAF sub-processes it implements. The SAF does not prescribe the order of sub-processes. GSBPM is not a strict sequence: sub-processes can be skipped, repeated or run in parallel.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Value streams (Strategy)	Processes realise value stream stages	PRD-4.3 *Run collection* realises VS-01.3 *Collect or Receive Data*
Capabilities (Strategy)	Processes realise capabilities	PRD-5.4 *Edit and impute* realises Adjusting and imputing (1.3.4)
Business functions	Processes realise functions; functions are assigned to processes	The production process realises the production function
Business roles	Roles are assigned to processes	ROL-14 *Data quality analyst* is assigned to PRD-5.3 *Review and validate*
Business events	Events trigger processes and mark their results	*Collection period opened* triggers PRD-4.3
Business objects	Processes access business objects	PRD-5.8 *Finalise data files* writes the processed microdata
Steady states	Processes produce and consume steady states	DLM-3 *Register and catalogue data* registers a dataset in a steady state
Application services	Application services serve processes	A validation service serves PRD-5.3

Process groups

Group	IDs	Basis	Scope
A. Statistical production	PRD-1 to PRD-8	GSBPM phases and sub-processes	The production of a statistical product, from need to evaluation
B. Overarching processes	OVA-1 to OVA-6	GSBPM overarching processes	Activities that run across all production phases
C. Data management and lifecycle	DLM-1 to DLM-11	SAF	The management of data assets and steady states across products
D. Microdata services	MDS-1 to MDS-9	SAF	Controlled access to microdata for research
E. Data spaces	DSP-1 to DSP-6	SAF	Participation in data spaces, as data provider and data user

PLACEHOLDER: ArchiMate view of the SAF process model

A. Statistical production processes

The statistical production processes follow the eight phases of GSBPM. Each phase is described first, followed by its sub-processes. The descriptions are paraphrased from GSBPM, a model of UNECE published under the Creative Commons Attribution 4.0 licence (CC BY 4.0).

PRD-1 Specify Needs

ID	Process	Description
PRD-1	**Specify Needs**	This phase starts when a new statistical need is identified, or when an existing product needs review. It establishes what users need and whether the NSI can meet that need. The phase explores options, including the use of data that already exists. It ends with a decision on whether to proceed, based on a business case. For existing products that run regularly, this phase is usually carried out only when a review is due.
PRD-1.1	Identify needs	This sub-process investigates which statistics are needed and what they are needed for. The trigger can be a new policy question, a legal requirement or a user request. It also looks at changes in the needs for existing statistics. Input comes from users, stakeholders, legislation and earlier evaluations. The result is a first description of the need.
PRD-1.2	Consult and confirm needs	This sub-process consults users and stakeholders to understand the need in detail. It checks what exactly they want to know, at which level of detail and how often. It also explores which constraints apply, such as timing or budget. The NSI and the users confirm the need together. The result is a confirmed and documented set of needs.
PRD-1.3	Establish output objectives	This sub-process translates the confirmed needs into output objectives. It defines which statistics will be produced and which quality they must meet. Legal and confidentiality constraints are taken into account. The objectives are realistic for the data and resources that are available. The result is the basis for the design of the outputs.

ID	Process	Description
PRD-1.4	Identify concepts	This sub-process clarifies the concepts to be measured, from the point of view of the users. A concept such as “household income” or “unemployment” needs a clear meaning. The sub-process checks whether existing statistical concepts can be reused. Concepts may not yet match the definitions that data sources use. That mismatch is resolved later, in the design phase.
PRD-1.5	Check data availability and suitability	This sub-process checks whether existing data sources can meet the need. It looks at internal data, administrative data, data spaces and other external sources. For each source, it assesses quality, coverage, timeliness and legal conditions. Reusing existing data comes before new collection, in line with SAF-P08. The result shows which gaps remain that new data must fill.
PRD-1.6	Prepare and submit business case	This sub-process documents the findings of the phase in a business case. The business case describes the need, the proposed solution, the costs and the benefits. It also lists risks and alternatives. The business case is submitted to management for a decision. Approval starts the design phase.

PRD-2 Design

ID	Process	Description
PRD-2	**Design**	This phase designs the statistical product and the process that produces it. It covers outputs, variables, collection, sampling, processing, analysis and dissemination. It also defines the metadata and the quality criteria. Designs reuse existing standards, methods and components wherever possible. The results are the specifications for the build phase.

ID	Process	Description
PRD-2.1	Design outputs and dissemination	This sub-process designs the statistical outputs in detail. It defines tables, datasets, indicators and their metadata. It also designs how and through which channels the outputs will be disseminated. Disclosure control requirements are taken into account from the start. The design follows the output objectives from PRD-1.3.
PRD-2.2	Design variables	This sub-process defines the variables to be collected and to be derived. It specifies their definitions, the types of statistical units and the classifications that apply. Existing variables and classifications are reused where possible. Each variable gets metadata that describes its meaning. The result is a variable list that all later phases use.
PRD-2.3	Design collection	This sub-process determines how the data will be collected. It chooses the sources, the collection modes and the instruments, such as questionnaires or data interfaces. For administrative data and data spaces, it designs the agreements and delivery formats. Respondent burden is kept as low as possible. The design also covers the metadata and paradata to be captured.
PRD-2.4	Design frame and sample	This sub-process defines the target population and the frame from which units are selected. It designs the sampling method and the sample size, where a sample is used. For register-based statistics, it defines how the population is derived from the registers. Coverage problems in the frame are identified and addressed. The result is the method used later in PRD-4.1.

ID	Process	Description
PRD-2.5	Design processing and analysis	This sub-process designs the methods for processing and analysis. It covers integration, coding, validation, editing, imputation, derivation, weighting and estimation. It also designs the methods for disclosure control. Methods are documented so that results can be reproduced. The designs are reviewed by methodologists before they are built.
PRD-2.6	Design production systems and workflows	This sub-process designs the end-to-end workflow of the statistical process. It defines which systems and components are needed for each step. It avoids gaps and duplication between steps. Reuse of existing components and shared platforms comes first, in line with SAF-P01. The result is the blueprint for the build phase.

PRD-3 Build

ID	Process	Description
PRD-3	**Build**	This phase builds and tests the production solution. It assembles the components, configures the workflows and tests the whole process. Components are reused where possible and built only where needed. The phase ends when the process is ready for production. For regular products, this phase is only needed when something changes.
PRD-3.1	Reuse or build collection instruments	This sub-process builds or reuses the instruments for data collection. Examples are electronic questionnaires, data delivery interfaces and connectors to data spaces. It follows the design from PRD-2.3. Instruments are tested for usability and correctness. The result is a set of instruments ready for collection.

ID	Process	Description
PRD-3.2	Reuse or build processing and analysis components	This sub-process builds or reuses the components for processing and analysis. Examples are validation rules, imputation routines and estimation programs. The components implement the methods designed in PRD-2.5. Shared components that other products already use are preferred. Each component is documented and version-controlled.
PRD-3.3	Reuse or build dissemination components	This sub-process builds or reuses the components for dissemination. Examples are table templates, publication workflows and API endpoints. It follows the design from PRD-2.1. Shared dissemination platforms are used where possible. The components are tested together with the output systems.
PRD-3.4	Configure workflows	This sub-process configures the workflows that connect the components. It sets up the order of steps, the data flows and the transformations. It also configures the registration of data in the steady states. Configuration is preferred over custom code. The result is an executable workflow.
PRD-3.5	Test production systems	This sub-process tests the assembled systems and their interactions. It checks that components work together and handle errors correctly. It also tests performance and security. Test results are documented and reported to the process owner. Defects are fixed before the process goes live.
PRD-3.6	Test statistical business process	This sub-process tests the whole statistical process in practice. It can take the form of a pilot or a field test with real data. It checks that the process produces results of the expected quality. Problems found in the test lead to changes in design or build. The process owner decides whether the process is ready.

ID	Process	Description
PRD-3.7	Finalise production systems	This sub-process puts the process and its systems into production. It completes the documentation and trains the staff involved. It transfers the systems to the teams that will run and maintain them. The metadata of the process is registered. From this moment, the process can run in the collect to disseminate phases.

PRD-4 Collect

ID	Process	Description
PRD-4	**Collect**	This phase collects or acquires the data, metadata and paradata needed for the product. It covers surveys, administrative data, data from data spaces and other sources. Data is received securely and checked for completeness. The phase ends when the data is stored and registered as raw data. In the steady-state model, this phase produces State #1.
PRD-4.1	Create frame and select sample	This sub-process creates the frame for this iteration of the process. It applies the method designed in PRD-2.4 to the current population. Where a sample is used, it draws the sample. The quality of the frame and the sample is checked. The result is the list of units from which data will be collected.
PRD-4.2	Set up collection	This sub-process makes sure that people, processes and systems are ready to collect. It prepares the instruments, trains interviewers and informs respondents or data providers. It checks that agreements and access rights are in place. Collection channels are tested before collection starts. The collection period is planned and communicated.
PRD-4.3	Run collection	This sub-process carries out the collection. Respondents complete questionnaires, and data providers deliver their data. Progress and response are monitored continuously. Non-response is followed up within the agreed strategy. Paradata about the collection is recorded.

ID	Process	Description
PRD-4.4	Finalise collection	This sub-process loads the collected data and metadata into a secure environment. It checks integrity, completeness and structure at intake. It registers the data as a new version of raw data. Delivery problems are reported to the data provider. The collection is closed when all expected data has been received or written off.

PRD-5 Process

ID	Process	Description
PRD-5	**Process**	This phase turns the collected data into microdata that is ready for analysis. It integrates, classifies, checks, corrects and derives data. It also calculates weights and aggregates. The phase runs across several business domains of the steady-state model. It produces States #2 and #3, and the first aggregates of State #4.
PRD-5.1	Integrate data	This sub-process combines data from one or more sources into a consistent dataset. It links records that describe the same unit, using agreed keys. It resolves overlaps and conflicts between sources. Linkage uses pseudonymised keys where personal data is involved. The result is an integrated dataset with documented linkage quality.
PRD-5.2	Classify and code	This sub-process assigns codes from standard classifications to the data. Examples are codes for economic activity, occupation or region. Coding can be automatic, manual or a mix of both. The classification versions used are recorded in the metadata. Coding quality is checked and documented.

ID	Process	Description
PRD-5.3	Review and validate	This sub-process checks the data for errors, outliers and inconsistencies. It applies the validation rules designed in PRD-2.5. It flags records that need attention. Some checks run automatically, others need expert review. The result is a dataset with quality flags and a list of issues.
PRD-5.4	Edit and impute	This sub-process corrects incorrect values and fills in missing values. Corrections follow editing rules, and missing values are imputed with documented methods. Every change is recorded, so that the original value can always be traced. The impact of editing and imputation on the results is measured. The result is a complete and consistent dataset.
PRD-5.5	Derive new variables and units	This sub-process creates variables and units that were not collected directly. Examples are age groups, household composition or enterprise groups. Derivations follow the specifications from PRD-2.2. Derived variables get their own metadata. The derivation rules are documented and reusable.
PRD-5.6	Calculate weights	This sub-process calculates weights for the records in the dataset. Weights correct for the sample design, non-response and coverage. They follow the method designed in PRD-2.5. The quality of the weights is checked. The weights are stored with the microdata.
PRD-5.7	Calculate aggregates	This sub-process calculates totals and other aggregates from the microdata. It applies the weights and the estimation methods. It can also combine lower-level aggregates into higher levels. Accuracy measures are calculated alongside the estimates. The result is the first version of the statistics.

ID	Process	Description
PRD-5.8	Finalise data files	This sub-process combines the results of the phase into the files used for analysis. It checks that the files are complete and consistent. It registers the processed microdata as a new version in the steady state. The metadata of the files is completed. The files are now ready for the analysis phase.

PRD-6 Analyse

ID	Process	Description
PRD-6	**Analyse**	This phase turns the processed data into statistical outputs and checks that they are fit for release. It prepares draft outputs, validates them and explains them. It also applies disclosure control. The phase ends with the approval of the content for release. In the steady-state model, it completes State #4.
PRD-6.1	Prepare draft outputs	This sub-process transforms the data files into draft statistical outputs. Examples are tables, indicators, time series and charts. It also prepares the methodological notes that go with them. Drafts follow the output design from PRD-2.1. They are the input for validation and interpretation.
PRD-6.2	Validate outputs	This sub-process checks the draft outputs for quality and plausibility. It compares results with earlier periods, other sources and expectations. It checks them against the quality framework. Unexpected results are investigated before they are accepted. The result is a set of validated outputs with documented quality.
PRD-6.3	Interpret and explain outputs	This sub-process interprets the outputs and explains what they mean. Subject-matter experts describe developments and their causes. Deeper analyses may be carried out to support the explanation. The interpretation must be neutral and based on the data. The result is the content for publications and press releases.

ID	Process	Description
PRD-6.4	Apply disclosure control	This sub-process makes sure that outputs do not reveal information about individual respondents. It applies the disclosure control methods designed in PRD-2.5. Examples are suppression of cells, rounding and adding noise. The disclosure risk is assessed and documented. No output is released without passing this check.
PRD-6.5	Finalise outputs	This sub-process makes sure that the statistics are fit for use. It completes the outputs and their metadata. It checks that all quality and confidentiality requirements are met. The content is approved for release by the responsible role. The approved outputs are registered as statistics ready for dissemination.

PRD-7 Disseminate

ID	Process	Description
PRD-7	**Disseminate**	This phase publishes the statistical products and supports users. It updates the output systems, produces the products and releases them. It also promotes the products and answers user questions. Equal access for all users at the moment of release is a core rule. In the steady-state model, this phase produces State #5.
PRD-7.1	Update output systems	This sub-process loads the approved data and metadata into the systems used for dissemination. Examples are databases behind the website, open data portals and APIs. It checks that the data is loaded correctly and remains embargoed until release. Format and structure follow the dissemination design. The result is output that is ready for release.

ID	Process	Description
PRD-7.2	Produce dissemination products	This sub-process produces the products that users receive. Examples are press releases, publications, tables, visualisations and datasets. It follows the design from PRD-2.1. Products are checked for correctness and consistency before release. The result is a set of products ready to publish.
PRD-7.3	Release dissemination products	This sub-process releases the products to users. It follows the release calendar and the embargo rules. All channels are updated at the same moment, so that no user gets an advantage. Errors found after release are corrected openly. The release is logged for accountability.
PRD-7.4	Promote dissemination products	This sub-process brings the products to the attention of users. It uses channels such as social media, newsletters and events. It targets the user segments that have an interest in the product. Promotion helps users find and understand the statistics. Its effect is monitored.
PRD-7.5	Provide user support	This sub-process answers questions and requests from users. Questions are registered and answered within agreed times. Frequent questions lead to better documentation or products. Requests for tailor-made data are handled according to confidentiality rules. Feedback from users is passed on to the product owners.

PRD-8 Evaluate

ID	Process	Description
PRD-8	**Evaluate**	This phase evaluates a specific run of the statistical process. It collects evidence, assesses quality and performance, and agrees on improvements. Evaluation can take place after every run or at set intervals. Its results feed back into the next run or into a redesign. The phase is part of the management process type of each business domain.

ID	Process	Description
PRD-8.1	Gather evaluation inputs	This sub-process collects the material for the evaluation. Examples are quality indicators, paradata, process logs, user feedback and staff comments. The inputs come from all earlier phases. They are brought together in one place. The result is a complete evidence base for the evaluation.
PRD-8.2	Conduct evaluation	This sub-process analyses the inputs and draws conclusions. It assesses the quality of the outputs and the efficiency of the process. It compares results with targets and with earlier runs. Strengths and weaknesses are identified. The result is an evaluation report or dashboard.
PRD-8.3	Agree an action plan	This sub-process agrees on actions to improve the process. It sets priorities, owners and deadlines. Actions can lead to small changes or to a full redesign. Progress on the actions is monitored. The action plan is the input for the next cycle.

B. Overarching processes

Overarching processes run across all phases of statistical production. They are not steps in the sequence. They apply to every phase, and to every product. They are based on the overarching processes of GSBPM.

ID	Process	Description
OVA-1	Quality Management	This process assesses and controls the quality of products and processes. It applies the quality framework of the NSI in every phase. It defines quality indicators, measures them and reports on them. Quality problems lead to corrective actions. The process supports the NSI-wide quality management function.

ID	Process	Description
OVA-2	Metadata Management	This process creates, updates, uses and archives statistical metadata in all phases. Metadata describes concepts, variables, classifications, methods and processes. It is captured when it is created, not afterwards. Metadata is shared through a catalogue, so that others can find and understand the data. The process applies SAF-P09 *No data without metadata and classification*.
OVA-3	Data Management	This process manages data throughout its life cycle in production. It covers storage, security, versioning, retention and disposal. It makes sure that data is available to the right people at the right time. The SAF details this process in group C, *Data management and lifecycle processes*. The process applies the design rules of the steady-state model.
OVA-4	Process Data Management	This process records and uses data about the execution of the statistical process. Examples are paradata from collection, logs of processing steps and timings. Process data shows how the process actually ran. It is used for monitoring, evaluation and audit. It is also the basis for lineage: the trace from output back to source.
OVA-5	Knowledge Management	This process maintains the knowledge needed to run the production process. It keeps documentation, methods and procedures up to date. It makes sure that recurring processes can be repeated in the same way. It also captures lessons learned from evaluations. Knowledge management reduces the dependence on individual staff members.

ID	Process	Description
OVA-6	Data Supplier Management	This process manages the technical and operational relationship with data suppliers during production. It covers delivery schedules, formats, quality feedback and problem resolution. It keeps track of the burden placed on suppliers. It works within the agreements made at corporate level. Good supplier management improves data quality and reduces delays.

C. Data management and lifecycle processes

These processes manage data as an asset, across products and over its whole life. They detail the overarching process OVA-3 *Data Management*. They follow the life cycle of a steady state: promotion, gate, registration, use, revision and retirement. They apply to every reference model for the data value chain.

ID	Process	Description
DLM-1	Plan and design data assets	This process decides which data assets the NSI needs and how they are structured. It defines datasets, their owners, their classification and their place in the steady states. It is triggered by a new product, a new source or a change in the data architecture. Planning prevents duplicate datasets and unclear ownership. The result is a documented design for each data asset.
DLM-2	Promote data through a quality gate	This process submits a dataset to the quality gate of a steady state. It checks the dataset against the gate criteria, such as completeness, structure and required metadata. The outcome is acceptance, conditional acceptance with quality flags, or rejection. Rejected data returns to the producing domain or the data provider. The process makes sure that only data of known quality enters a steady state.

ID	Process	Description
DLM-3	Register and catalogue data	This process registers an accepted dataset as a new, immutable version. It assigns a persistent identifier and records the metadata in the catalogue. It publishes an event that tells consuming domains a new version is available. Registration makes the data findable for authorised users. Without registration, data cannot be handed over.
DLM-4	Classify and protect data	This process assigns a security classification to each dataset, for availability, integrity and confidentiality. It applies the protection measures that follow from the classification. It also applies pseudonymisation or other privacy measures where required. Classification is reviewed when data moves to the next steady state. The process applies SAF-P09, SAF-P10 and SAF-P11.
DLM-5	Store and version data	This process stores each registered version of a dataset safely. Earlier versions are kept, not overwritten. Backups and recovery are arranged according to the availability requirements. Storage follows the retention policy of each steady state. The process makes it possible to reproduce any statistic from the data it was built on.
DLM-6	Provide data for reuse	This process gives authorised internal users access to registered data. A user requests access, and the data owner decides on the request. Access is granted for a stated purpose and a set period. Use is logged. The process applies SAF-P08 *Data is shared property*, within the limits of confidentiality.
DLM-7	Revise data	This process handles corrections and redeliveries of data that is already registered. A correction creates a new version that passes the same quality gate. Consumers of the earlier version are notified. They decide whether to reprocess their own data. The reason for each revision is recorded.

ID	Process	Description
DLM-8	Monitor data quality, lineage and use	This process monitors registered data over time. It tracks quality indicators, the lineage from source to output, and how the data is used. It detects problems, such as declining quality or unauthorised use. It reports to data owners and the data governance board. The results feed into improvements and audits.
DLM-9	Manage reference and master data	This process manages the data that other data depends on. Examples are classifications, code lists, statistical registers and population frames. It controls versions and publishes changes in advance. It makes sure that all domains use the same version at the same time. Errors in reference data affect many products, so changes are tested carefully.
DLM-10	Archive and preserve data	This process transfers data that must be kept long-term to an archive. It selects what must be preserved, based on legal obligations and reproducibility needs. It makes sure that archived data remains readable and understandable, with its metadata. Access to archived data is controlled. The process follows national archives legislation.
DLM-11	Dispose of data	This process deletes data at the end of its retention period. It checks that no statistic or obligation still depends on the data. Deletion covers all copies, including backups and working copies. Keys used for pseudonymisation are destroyed when they are no longer needed. Each disposal is recorded, so that it can be shown in an audit.

D. Microdata services processes

Microdata services give researchers controlled access to microdata for statistical and scientific research. Access takes place under strict conditions, usually in a secure processing environment. The NSI remains responsible for confidentiality throughout. These processes realise part of VS-06 *Dissemination and Service Delivery* and work within the collaboration COL-04 *Research access partnership*.

ID	Process	Description
MDS-1	Authorise institutions and researchers	This process decides which institutions and researchers may use microdata services. It checks the legal status of the institution and the qualifications of the researchers. Authorised institutions sign an agreement that sets the conditions of use. Researchers confirm that they will respect confidentiality. Authorisations are reviewed periodically and can be withdrawn.
MDS-2	Assess access applications	This process handles an application to use microdata for a specific research project. The applicant describes the purpose, the data needed and the expected outputs. The NSI checks the legal basis, the public interest and whether the data requested is necessary. It also checks for special categories of data that need extra approval. The result is an approval with conditions, or a reasoned refusal.
MDS-3	Prepare research datasets	This process prepares the datasets for an approved project. It selects only the data that the project needs. It replaces identifiers with project-specific pseudonyms, so that data cannot be linked across projects. Where the project brings its own data, the NSI links it under the same rules. The prepared dataset is registered with its metadata and lineage.
MDS-4	Provide a secure processing environment	This process sets up the environment in which researchers work with the data. The environment prevents data from leaving without control. Access is limited to the authorised researchers of the project. Tools and computing power are provided as agreed. All activity in the environment is logged.

ID	Process	Description
MDS-5	Facilitate privacy-preserving analysis	This process supports analysis without moving the data into one place. Examples are federated analysis, where the algorithm goes to the data, and secure multi-party computation. These techniques are used when partners cannot or may not share their data. The NSI defines the trust model, the linkage method and the output rules for each setup. The process extends microdata services to data that the NSI does not hold itself.
MDS-6	Support researchers	This process helps researchers use the data and the environment correctly. It answers questions about content, metadata and methods. It explains the rules for confidentiality and output. It handles technical problems in the environment. Good support reduces errors and speeds up research.
MDS-7	Check and release output	This process checks every output before it leaves the secure environment. It applies disclosure control rules to tables, models and other results. It also checks for risks across outputs, such as differencing between similar results. Outputs that pass are released to the researcher. Outputs that fail are returned with an explanation.
MDS-8	Monitor use and compliance	This process monitors whether institutions and researchers follow the conditions of use. It reviews logs, outputs and incidents. It also tracks the use of query or disclosure budgets where these apply. Breaches lead to measures, up to withdrawal of the authorisation. The results are reported to the data owner and to supervisory authorities where required.

ID	Process	Description
MDS-9	Close projects	This process ends a research project in a controlled way. It removes the project's data and workspace. It destroys the project-specific pseudonymisation keys. It archives what must be kept for reproducibility, such as the code and the approved outputs. The closure is recorded, so that it can be shown in an audit.

E. Data space processes

A data space is a shared environment in which organisations exchange data under common rules, while each keeps control over its own data. Examples are the European data spaces for health, mobility or public administration. An NSI can take part as a data provider, as a data user, or both. Some NSIs also act as an access body that grants access to data on behalf of other data holders. That role is not covered by the SAF reference processes. The data space processes realise parts of VS-01 *Data Acquisition and Ingestion* and VS-06 *Dissemination and Service Delivery*.

ID	Process	Description
DSP-1	Join a data space	This process prepares the NSI to take part in a data space. It assesses the rules of the data space against the NSI's legal mandate and policies. It signs the participation agreement and obtains the required identity and trust credentials. It sets up the technical connection, for example through a connector. The NSI decides in advance whether it joins as provider, as user, or as both.
DSP-2	Publish data offerings	This process makes NSI data visible in the data space. It describes datasets in the data space catalogue, using agreed metadata standards. It states the conditions under which the data may be used. Only data that the NSI may share under its legal mandate is offered. The offering is kept up to date when the data or the conditions change.

ID	Process	Description
DSP-3	Agree data use conditions	This process agrees on the terms of a specific data exchange. Provider and user agree on purpose, duration, permitted use and security measures. The terms are recorded in a contract or usage policy that machines can check where possible. The NSI checks that the terms fit its confidentiality rules. Without agreed terms, no data is exchanged.
DSP-4	Exchange data under usage control	This process carries out the agreed data exchange. Data is transferred securely, or analysed where it resides without transfer. Usage control enforces the agreed conditions where the technology allows it. Every exchange is logged. Incoming data enters the NSI through the data acquisition process and is registered as raw data.
DSP-5	Monitor compliance and report	This process monitors whether the NSI's own exchanges follow the rules of the data space. It checks agreements, logs, incidents and the use of data. It reports security incidents within the legal deadlines. It also checks whether partners respect the conditions agreed for NSI data. Breaches lead to measures, including suspension of the exchange.
DSP-6	Participate in data space governance	This process covers the NSI's role in steering the data space. The NSI takes part in governance bodies and contributes to the rules and standards. It brings in statistical expertise on metadata, classifications, quality and confidentiality. It also evaluates whether participation still serves its mandate. When the NSI leaves a data space, this process arranges a controlled exit.

PLACEHOLDER: Process templates and process mapping instruments to be added to the toolkit

Business Functions

The concept

A **business function** is a grouping of behaviour based on the knowledge, skills and resources it needs. A function describes *what* the organisation does on a continuous basis. It has no start or end, and it does not depend on the order in which work is done. *Manage metadata* is a function: it exists as long as the NSI exists, whatever process happens to use it. Functions are stable, because the competences an organisation needs change much more slowly than its procedures.

Functions differ from processes. A process is a sequence of activities with a trigger and a result. A function is a lasting area of competence that processes draw on. One process uses several functions, and one function serves several processes. The mapping at the end of this section shows how they relate.

Functions also differ from capabilities. A capability, in the Strategy layer, describes an *ability* the NSI has or needs. A function, in the business layer, describes the *behaviour* that delivers that ability. Functions realise capabilities. In practice, the two often have similar names, but they serve different purposes: capabilities support planning and investment, functions support the design of the organisation and its processes.

How the SAF applies business functions

Why the SAF describes business functions

Functions give a stable picture of the organisation that does not change with every reorganisation or new system. They are the natural basis for assigning responsibilities: each function has an owner who maintains the competence. They also help to compare NSIs. Two NSIs may have very different processes and departments, but they will recognise the same functions. Finally, functions make gaps visible. A function that no one owns is a competence that the organisation risks losing.

How business functions fit into an NSI's own architecture

An NSI maps its departments and teams onto the SAF functions. The mapping shows which unit carries which function, and where functions are spread over several units. An NSI can also map the SAF functions onto a national function model, such as a government-wide model for business support. Functions that are specific to the NSI's mandate are added as extensions. The SAF function areas for statistical production, data management, microdata services and data spaces are specific to official statistics. The business support area is generic and can be replaced by a national model where one exists.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Capabilities (Strategy)	Functions realise capabilities	FUN-7.2 *Manage metadata* realises Metadata management (2.1.2)
Business processes	Processes realise functions; functions are assigned to processes	PRD-5.4 *Edit and impute* uses FUN-3.3 *Process statistical data*
Business roles	Roles are assigned to functions	ROL-03 *Data steward* is assigned to FUN-7.3 *Manage data lifecycle and storage*
Business actors	Organisational units carry functions	A methodology department carries FUN-4.1
Business services	Functions realise business services	FUN-8.1 realises the microdata access service
Application services	Application services serve functions	A metadata catalogue serves FUN-7.2

Function areas

Area	Name	Scope
FUN-1	Strategy and Leadership	Direction, governance and strategic relations of the NSI
FUN-2	Capability Development	Planning, developing and maintaining the abilities of the NSI, including architecture
FUN-3	Statistical Production	Designing, producing and disseminating statistics
FUN-4	Methodology and Research	Statistical methods, research and innovation
FUN-5	Quality Management	The quality framework and quality assurance
FUN-6	Data Supplier Management	Relations, agreements and burden for data providers and respondents
FUN-7	Data Management	Data and metadata as managed assets across their life cycle
FUN-8	Microdata Services	Controlled access to microdata for research
FUN-9	Data Space Participation	Taking part in data spaces as provider and user
FUN-10	Business Support	The generic support functions of the organisation

PLACEHOLDER: ArchiMate view of the SAF function model

FUN-1 Strategy and Leadership

ID	Function	Description	Capabilities
FUN-1	**Strategy and Leadership**	This area gives direction to the NSI. It sets the vision, the strategy and the priorities. It governs the organisation and accounts for its results. It maintains relationships with the government, the statistical community and society. All other functions work within the direction this area sets.	4.1

ID	Function	Description	Capabilities
FUN-1.1	Define vision and strategy	This function defines the mission, vision, goals and values of the NSI. It analyses the national and international environment, including the drivers in the Motivation layer. It translates the vision into a strategy and domain strategies. It sets priorities for the product and service portfolio. The results guide investment and change for several years.	4.1.1, 4.1.2
FUN-1.2	Govern and steer the organisation	This function makes sure the NSI is run according to its strategy and its legal mandate. It sets up governance bodies, decision rights and the planning and control cycle. It allocates resources to programmes and departments. It monitors performance and intervenes when needed. It also accounts to the ministry, supervisory bodies and the public.	4.1.1, 4.1.5
FUN-1.3	Manage strategic collaboration	This function maintains the strategic relationships of the NSI with other organisations. It covers the national statistical system, international statistical organisations and other governments. It identifies opportunities for joint work, data exchange and shared solutions. It represents the NSI in working groups and governance bodies. It makes sure the NSI speaks with one voice.	4.1.6

ID	Function	Description	Capabilities
FUN-1.4	Build public confidence	This function builds and protects public trust in official statistics. It explains the role and independence of the NSI to the public. It promotes the value of statistics and of providing data to the NSI. It responds to misuse or misinterpretation of statistics. Trust is the basis for both response and use, so this function supports the whole chain.	1.5.2, 4.1.5

FUN-2 Capability Development

ID	Function	Description	Capabilities
FUN-2	**Capability Development**	This area develops and improves the abilities of the NSI. It plans improvements, carries them out and checks whether they deliver. It covers methods, processes, data, systems and skills. It also maintains the enterprise architecture that keeps improvements coherent. This area turns the strategy into concrete change.	4.1.3, 4.1.4
FUN-2.1	Plan capability improvements	This function identifies which capabilities need to improve and in which order. It uses capability assessments, evaluations and strategic priorities. It proposes programmes and projects with a business case. It coordinates the plans across the organisation to avoid overlap. The result is a prioritised portfolio of improvements.	4.1.3

ID	Function	Description	Capabilities
FUN-2.2	Develop capability improvements	This function carries out approved improvements. It specifies, designs, builds and tests the change. It follows the architecture principles and reuses existing solutions where possible. It involves the people who will use the result. The function ends when the improvement is ready to be taken into use.	4.1.3, 3.2.1
FUN-2.3	Monitor capabilities	This function checks whether capabilities perform as intended. It measures maturity and the benefits of past investments. It identifies capabilities that are declining or no longer needed. It reports to management and feeds the next planning round. Without monitoring, investments cannot be shown to pay off.	4.1.4
FUN-2.4	Transfer capability improvements to operations	This function hands over a finished improvement to the teams that will run and support it. It arranges documentation, training and support agreements. It checks that the operating teams are ready before the handover. It closes the project or programme. A clean transfer prevents improvements from falling back into project mode.	3.2.2

ID	Function	Description	Capabilities
FUN-2.5	Manage enterprise architecture	This function develops and maintains the enterprise architecture of the NSI. It uses the SAF as its reference. It advises on strategy, investments and projects, and reviews them against the architecture principles. It supports the architecture board and keeps the architecture repository up to date. It keeps the NSI architecture aligned with the SAF as both evolve.	

FUN-3 Statistical Production

ID	Function	Description	Capabilities
FUN-3	**Statistical Production**	This area produces official statistics, from the first design to the release. It is the core of the NSI's mandate. The statistical production processes (PRD) realise this area. Its functions follow the main kinds of statistical work: design, collection, processing, estimation, confidentiality and dissemination. Each function brings together the specialist knowledge for that kind of work.	1
FUN-3.1	Design and build statistical products	This function designs statistical products and the processes that produce them. It translates information needs into outputs, variables, methods and workflows. It also builds and tests the production solution. It works closely with methodology and with the users of the product. The function owns the design of each product over its life cycle.	1.1

ID	Function	Description	Capabilities
FUN-3.2	Collect and acquire data	This function obtains the data that statistical products need. It runs surveys, receives administrative data and acquires data from other sources, including data spaces. It makes sure that data arrives complete, on time and under the right agreements. It keeps respondent burden as low as possible. The function delivers data into the first steady state.	1.2
FUN-3.3	Process statistical data	This function turns raw data into reliable microdata. It integrates, standardises, codes, validates, edits and imputes data. It also derives new variables and units. It applies the methods designed for each product. The function produces standardized and processed microdata.	1.3.1, 1.3.2, 1.3.3, 1.3.4
FUN-3.4	Estimate and analyse	This function produces statistics from microdata and checks that they are sound. It calculates weights, estimates and aggregates. It integrates statistics into consistent wholes, such as the national accounts. It validates and interprets the results. The function produces the statistics that are ready for release.	1.3.5, 1.3.6, 1.4

ID	Function	Description	Capabilities
FUN-3.5	Protect statistical confidentiality	This function makes sure that no output reveals information about individual respondents. It sets the confidentiality rules and the methods for disclosure control. It applies them to tables, microdata files and services. It assesses disclosure risks in new types of output. The function works across all products and channels.	1.3.7
FUN-3.6	Disseminate statistics	This function brings statistics to users. It manages the release calendar, the publication channels and the products. It makes sure that all users get access at the same moment. It also supports users in finding and understanding the statistics. The function produces released data.	1.5

FUN-4 Methodology and Research

ID	Function	Description	Capabilities
FUN-4	**Methodology and Research**	This area provides the scientific basis for statistical production. It develops and maintains statistical methods. It explores new data sources, new techniques and new products. It makes sure that innovations are sound before they enter production. This area protects the methodological quality that makes official statistics credible.	2.2

ID	Function	Description	Capabilities
FUN-4.1	Manage statistical and analytical methodology	This function develops, documents and maintains the methods used in production. It covers sampling, collection, editing, imputation, estimation, disclosure control and analysis. It includes methods based on machine learning and AI. It advises production teams and reviews their methods. Methods are standardised and shared across products where possible.	2.2.3
FUN-4.2	Research and innovate	This function explores new possibilities for statistics. It investigates new data sources and new ways to use existing data. It develops prototypes of new products and methods. It evaluates whether innovations are feasible and sound. Successful innovations are handed over to production through capability development.	2.2.1, 2.2.2

FUN-5 Quality Management

ID	Function	Description	Capabilities
FUN-5	**Quality Management**	This area makes sure that the NSI delivers statistics of the required quality. It maintains the quality framework and the tools to apply it. It assures quality in products and processes. It prepares the NSI for reviews and audits. This area is the corporate counterpart of the quality management process in production.	2.1.3

ID	Function	Description	Capabilities
FUN-5.1	Manage the quality framework	This function sets the quality standards of the NSI. It defines quality dimensions, indicators and reporting requirements. It aligns the framework with international standards, such as the European Statistics Code of Practice. It provides tools and guidance to production teams. The framework is reviewed and improved regularly.	2.1.3
FUN-5.2	Assure and review quality	This function checks whether products and processes meet the quality standards. It monitors quality indicators and reviews quality reports. It carries out quality reviews and self-assessments. It follows up on findings with improvement actions. It also supports external reviews, such as peer reviews.	2.1.3, 4.1.5

FUN-6 Data Supplier Management

ID	Function	Description	Capabilities
FUN-6	**Data Supplier Management**	This area manages the relationships with everyone who supplies data to the NSI. It covers respondents, administrative data holders and other data providers. It sets up and maintains agreements. It monitors the burden that the NSI places on suppliers. Good supplier management improves both the quality of data and the willingness to provide it.	3.4

ID	Function	Description	Capabilities
FUN-6.1	Manage data supplier relationships	This function maintains contact with data providers and respondent groups. It explains why the NSI needs the data and how it is used. It handles questions, complaints and feedback on data quality. It coordinates all contacts, so that a supplier deals with one NSI and not with many departments. The function builds long-term cooperation.	3.4.2
FUN-6.2	Manage data provision agreements	This function draws up, maintains and enforces agreements with data providers. Agreements cover content, format, timing, quality, security and legal basis. The function keeps a register of all agreements. It renews or adjusts agreements when sources or needs change. It also handles enforcement when obligations are not met.	3.4.3
FUN-6.3	Manage response burden	This function measures and limits the burden of statistical data collection. It tracks how much time and effort respondents and data providers spend. It looks for ways to reduce burden, such as reusing existing data. It coordinates surveys so that the same units are not approached too often. It reports on burden to management and government.	1.2, 2.2.5

FUN-7 Data Management

ID	Function	Description	Capabilities
FUN-7	**Data Management**	This area manages data and metadata as assets of the NSI. It covers governance, metadata, the data life cycle, reference data and data protection. It works across all statistical products and domains. It makes sure that data can be found, understood, trusted and reused. The data lifecycle management processes (DLM) realise this area.	2.1
FUN-7.1	Govern data	This function sets the rules for data management in the NSI. It defines data policies, standards and responsibilities. It assigns data owners, data stewards and data custodians. It resolves issues that cross domain boundaries. It monitors compliance with the data policies.	2.1.1, 4.1.2
FUN-7.2	Manage metadata	This function captures, maintains and publishes metadata. Metadata describes concepts, variables, classifications, datasets, methods and processes. The function maintains the catalogue in which data can be found. It makes sure that metadata is captured when data is created. It applies international standards, such as GSIM and SDMX.	2.1.2

ID	Function	Description	Capabilities
FUN-7.3	Manage data lifecycle and storage	This function manages data from creation to disposal. It covers quality gates, registration, versioning, storage, retention and disposal. It keeps track of lineage, from output back to source. It makes sure that earlier versions remain available when they are needed. The function applies the design rules of the steady-state model.	2.1.1
FUN-7.4	Manage reference and master data	This function manages the data that other data depends on. Examples are classifications, code lists, statistical registers and population frames. It controls versions and announces changes in advance. It makes sure that all domains use the same version. The function is the single source of truth for reference data.	2.1.4
FUN-7.5	Protect data	This function protects data according to its classification. It applies measures for availability, integrity and confidentiality. It manages pseudonymisation and the keys used for it. It keeps keys separate from the data they protect. The function works closely with information security and privacy management.	3.3.1, 3.3.4

FUN-8 Microdata Services

ID	Function	Description	Capabilities
FUN-8	**Microdata Services**	This area gives researchers controlled access to microdata. It covers authorisation, data preparation, secure environments and output control. It also supports analysis without moving data, using privacy-preserving techniques. The NSI remains responsible for confidentiality throughout. The microdata services processes (MDS) realise this area.	2.2.4
FUN-8.1	Manage research access	This function decides who may use microdata, for which purpose and under which conditions. It authorises institutions and researchers. It assesses applications for research projects. It monitors whether users keep to the conditions. It withdraws access when rules are broken.	2.2.4, 3.4.3
FUN-8.2	Prepare research data	This function prepares the datasets for each approved project. It selects only the data that the project needs. It applies project-specific pseudonyms, so that data cannot be linked across projects. It links data that researchers bring in, under the same rules. It documents the prepared data with metadata and lineage.	2.2.4, 3.3.1
FUN-8.3	Operate secure research environments	This function provides and runs the environments in which researchers work. It controls who may enter and what may leave. It provides tools and computing power. It logs all activity. It removes data and workspaces when projects end.	2.2.4, 3.2.5

ID	Function	Description	Capabilities
FUN-8.4	Control research output	This function checks all results before they leave the secure environment. It applies disclosure control rules to tables, models and other output. It checks for risks across outputs, such as differencing between similar results. It keeps a record of every release. The function protects respondents while allowing research to proceed.	1.3.7
FUN-8.5	Enable privacy-preserving analysis	This function makes analysis possible without bringing all data together. It uses techniques such as federated analysis and secure multi-party computation. It defines the trust model, the linkage method and the output rules for each setup. It works with partners who cannot or may not share their data. The function extends microdata services beyond the data the NSI holds itself.	2.2.4, 2.2.3

FUN-9 Data Space Participation

ID	Function	Description	Capabilities
FUN-9	**Data Space Participation**	This area enables the NSI to take part in data spaces. The NSI can act as a data provider, as a data user, or both. The area covers agreements, technical connection and governance. It makes sure that participation fits the NSI's legal mandate and confidentiality rules. The data space processes (DSP) realise this area.	4.1.6, 3.4

ID	Function	Description	Capabilities
FUN-9.1	Manage data space participation	This function decides in which data spaces the NSI takes part, and in which role. It assesses the rules of each data space against the NSI's mandate. It manages participation agreements, data offerings and usage conditions. It monitors whether exchanges follow the agreed rules. It reviews participation regularly.	4.1.6, 3.4.3
FUN-9.2	Operate data space connectivity	This function provides the technical connection to data spaces. It runs the connectors, identity credentials and catalogue entries. It carries out data exchanges under the agreed usage conditions. It logs every exchange. Incoming data is passed on to data collection and enters the first steady state.	3.4.1, 3.2.5
FUN-9.3	Contribute to data space governance	This function represents the NSI in the governance of data spaces. It contributes to rules, standards and rulebooks. It brings in statistical expertise on metadata, classifications, quality and confidentiality. It aligns positions with other NSIs where possible. It also arranges a controlled exit when participation ends.	4.1.6

FUN-10 Business Support

ID	Function	Description	Capabilities
FUN-10	**Business Support**	This area provides the people, money, information, technology and facilities that the other functions need. Its functions are similar in every public organisation. The SAF describes them only at a high level. An NSI can replace this area with the function model of its own government. The functions are still part of the SAF, because statistical work depends on them.	3
FUN-10.1	Manage human resources	This function makes sure the NSI has enough qualified staff. It covers recruitment, development, performance and the departure of staff. It plans the skills the NSI will need in the future. It supports managers in leading their teams. Skills such as data science and methodology are scarce, which makes this function critical.	3.1.1
FUN-10.2	Manage finance and procurement	This function manages the budget, the accounts and the purchase of goods and services. It plans and controls spending. It runs procurement procedures and manages supplier contracts. It reports on the financial position to management and government. It makes sure that spending follows the rules for public money.	3.1.2, 3.1.3

ID	Function	Description	Capabilities
FUN-10.3	Manage information technology	This function provides and manages the IT the NSI needs. It covers applications, platforms, infrastructure and the digital workplace. It develops, configures and maintains systems. It runs IT services and resolves disruptions. It plans the evolution of the technical landscape.	3.2
FUN-10.4	Manage information and records	This function manages the information created in management and support processes. It keeps records according to archives legislation. It makes information findable and accessible over time. It also manages the knowledge base of the organisation. Statistical data itself is managed by FUN-7 *Data Management*.	3.1.8
FUN-10.5	Manage legal affairs	This function provides legal advice and support. It interprets the statistics law, data protection law and other rules. It assesses agreements and contracts. It handles legal procedures and objections. It keeps the overview of legal obligations, for example in a legal register.	3.1.7
FUN-10.6	Manage communication	This function manages internal and external communication. It covers press relations, the website, social media and internal information. It makes sure the NSI communicates clearly and consistently. It supports the release of statistics with press releases and explanations. It also handles questions from the media.	3.1.4

ID	Function	Description	Capabilities
FUN-10.7	Manage facilities	This function provides the buildings, workplaces and physical services the NSI needs. It covers housing, maintenance, catering and physical access. It makes sure that buildings and spaces are safe. It supports field staff with the means they need. It plans facilities in line with ways of working, such as hybrid work.	3.1.6, 3.3.5
FUN-10.8	Manage information security and privacy	This function protects the NSI's information and the privacy of the people it concerns. It sets security and privacy policies and monitors compliance. It manages incidents and leads recovery. It assesses risks in new solutions and suppliers. It includes the role of the data protection officer.	3.3
FUN-10.9	Manage users and services	This function manages the relationship with the users of statistics. It identifies user groups and their needs. It manages the portfolio of products and services and their service levels. It handles questions, requests and complaints. It measures user satisfaction and passes feedback on to production.	3.4.1, 3.4.2, 2.2.5

Process and function mapping

The table shows which functions each process uses. The *main function* carries the knowledge the process mainly depends on. *Supporting functions* contribute specific expertise or resources. In ArchiMate, the process realises part of the main function, and the supporting functions are related through assignment or serving relationships.

PLACEHOLDER: ArchiMate view of the process and function mapping

A. Statistical production processes

Process	Name	Main function	Supporting functions
PRD-1	**Specify Needs**	FUN-3.1	FUN-10.9
PRD-1.1	Identify needs	FUN-3.1	FUN-10.9
PRD-1.2	Consult and confirm needs	FUN-3.1	FUN-10.9
PRD-1.3	Establish output objectives	FUN-3.1	FUN-5.1
PRD-1.4	Identify concepts	FUN-3.1	FUN-4.1
PRD-1.5	Check data availability and suitability	FUN-3.1	FUN-6.1, FUN-7.2
PRD-1.6	Prepare and submit business case	FUN-3.1	FUN-1.2, FUN-2.1
PRD-2	**Design**	FUN-3.1	FUN-4.1
PRD-2.1	Design outputs and dissemination	FUN-3.1	FUN-3.5, FUN-3.6
PRD-2.2	Design variables	FUN-3.1	FUN-7.2, FUN-7.4
PRD-2.3	Design collection	FUN-3.1	FUN-3.2, FUN-6.1
PRD-2.4	Design frame and sample	FUN-3.1	FUN-4.1, FUN-7.4
PRD-2.5	Design processing and analysis	FUN-4.1	FUN-3.1, FUN-3.5
PRD-2.6	Design production systems and workflows	FUN-3.1	FUN-2.5, FUN-10.3
PRD-3	**Build**	FUN-3.1	FUN-10.3
PRD-3.1	Reuse or build collection instruments	FUN-3.1	FUN-3.2
PRD-3.2	Reuse or build processing and analysis components	FUN-3.1	FUN-4.1, FUN-10.3
PRD-3.3	Reuse or build dissemination components	FUN-3.1	FUN-3.6, FUN-10.3
PRD-3.4	Configure workflows	FUN-3.1	FUN-7.3, FUN-10.3
PRD-3.5	Test production systems	FUN-3.1	FUN-10.3
PRD-3.6	Test statistical business process	FUN-3.1	FUN-5.2
PRD-3.7	Finalise production systems	FUN-3.1	FUN-2.4, FUN-7.2
PRD-4	**Collect**	FUN-3.2	FUN-6.1
PRD-4.1	Create frame and select sample	FUN-3.2	FUN-4.1, FUN-7.4
PRD-4.2	Set up collection	FUN-3.2	FUN-6.2
PRD-4.3	Run collection	FUN-3.2	FUN-6.1, FUN-6.3
PRD-4.4	Finalise collection	FUN-3.2	FUN-7.3
PRD-5	**Process**	FUN-3.3	FUN-3.4
PRD-5.1	Integrate data	FUN-3.3	FUN-7.4, FUN-7.5
PRD-5.2	Classify and code	FUN-3.3	FUN-7.4
PRD-5.3	Review and validate	FUN-3.3	FUN-5.2
PRD-5.4	Edit and impute	FUN-3.3	FUN-4.1
PRD-5.5	Derive new variables and units	FUN-3.3	FUN-7.2
PRD-5.6	Calculate weights	FUN-3.4	FUN-4.1
PRD-5.7	Calculate aggregates	FUN-3.4	FUN-4.1
PRD-5.8	Finalise data files	FUN-3.3	FUN-7.3
PRD-6	**Analyse**	FUN-3.4	FUN-3.5
PRD-6.1	Prepare draft outputs	FUN-3.4	FUN-3.6
PRD-6.2	Validate outputs	FUN-3.4	FUN-5.2

Process	Name	Main function	Supporting functions
PRD-6.3	Interpret and explain outputs	FUN-3.4	FUN-10.6
PRD-6.4	Apply disclosure control	FUN-3.5	FUN-4.1
PRD-6.5	Finalise outputs	FUN-3.4	FUN-5.2
PRD-7	**Disseminate**	FUN-3.6	FUN-10.9
PRD-7.1	Update output systems	FUN-3.6	FUN-10.3
PRD-7.2	Produce dissemination products	FUN-3.6	FUN-10.6
PRD-7.3	Release dissemination products	FUN-3.6	FUN-1.4
PRD-7.4	Promote dissemination products	FUN-3.6	FUN-10.6, FUN-1.4
PRD-7.5	Provide user support	FUN-10.9	FUN-3.6
PRD-8	**Evaluate**	FUN-5.2	FUN-2.3
PRD-8.1	Gather evaluation inputs	FUN-5.2	FUN-7.3
PRD-8.2	Conduct evaluation	FUN-5.2	FUN-2.3
PRD-8.3	Agree an action plan	FUN-2.1	FUN-5.2

B. Overarching processes

Process	Name	Main function	Supporting functions
OVA-1	Quality Management	FUN-5.2	FUN-5.1
OVA-2	Metadata Management	FUN-7.2	FUN-7.4
OVA-3	Data Management	FUN-7.3	FUN-7.1
OVA-4	Process Data Management	FUN-7.3	FUN-5.2
OVA-5	Knowledge Management	FUN-10.4	FUN-4.1
OVA-6	Data Supplier Management	FUN-6.1	FUN-6.3

C. Data lifecycle management processes

Process	Name	Main function	Supporting functions
DLM-1	Plan and design data assets	FUN-7.1	FUN-2.5
DLM-2	Promote data through a quality gate	FUN-7.3	FUN-5.2
DLM-3	Register and catalogue data	FUN-7.2	FUN-7.3
DLM-4	Classify and protect data	FUN-7.5	FUN-10.8
DLM-5	Store and version data	FUN-7.3	FUN-10.3
DLM-6	Provide data for reuse	FUN-7.1	FUN-7.3
DLM-7	Revise data	FUN-7.3	FUN-7.2
DLM-8	Monitor data quality, lineage and use	FUN-7.1	FUN-5.2
DLM-9	Manage reference and master data	FUN-7.4	FUN-7.2
DLM-10	Archive and preserve data	FUN-10.4	FUN-7.3
DLM-11	Dispose of data	FUN-7.3	FUN-7.5

D. Microdata services processes

Process	Name	Main function	Supporting functions
MDS-1	Authorise institutions and researchers	FUN-8.1	FUN-10.5
MDS-2	Assess access applications	FUN-8.1	FUN-10.5, FUN-7.1

Process	Name	Main function	Supporting functions
MDS-3	Prepare research datasets	FUN-8.2	FUN-7.5
MDS-4	Provide a secure processing environment	FUN-8.3	FUN-10.3, FUN-10.8
MDS-5	Facilitate privacy-preserving analysis	FUN-8.5	FUN-4.1, FUN-7.5
MDS-6	Support researchers	FUN-8.3	FUN-10.9
MDS-7	Check and release output	FUN-8.4	FUN-3.5
MDS-8	Monitor use and compliance	FUN-8.1	FUN-10.8
MDS-9	Close projects	FUN-8.3	FUN-7.3, FUN-7.5

E. Data space processes

Process	Name	Main function	Supporting functions
DSP-1	Join a data space	FUN-9.1	FUN-10.5, FUN-10.8
DSP-2	Publish data offerings	FUN-9.1	FUN-7.2
DSP-3	Agree data use conditions	FUN-9.1	FUN-10.5
DSP-4	Exchange data under usage control	FUN-9.2	FUN-3.2, FUN-7.3
DSP-5	Monitor compliance and report	FUN-9.1	FUN-10.8
DSP-6	Participate in data space governance	FUN-9.3	FUN-1.3

Functions without a reference process

Six functions are not used by any SAF reference process. They are carried out through management, change or support processes that are specific to each NSI. Many other functions appear only as supporting functions, which is normal for strategic and support functions.

Function	Carried out through
FUN-1.1 Define vision and strategy	The strategic planning cycle of the NSI
FUN-2.2 Develop capability improvements	Change programmes and projects
FUN-4.2 Research and innovate	Innovation projects (see VS-08)
FUN-10.1 Manage human resources	Generic business support processes
FUN-10.2 Manage finance and procurement	Generic business support processes
FUN-10.7 Manage facilities	Generic business support processes

Mapping to value stream stages

The mapping connects the Strategy layer to the business layer. A value stream stage describes the value that is added. The processes describe the work that adds it. The functions describe the competences that the work depends on.

PLACEHOLDER: ArchiMate view of value stream stages, processes and functions

Primary value streams

Stage	Name	Realised by processes	Main functions
VS-01.1	Identify Data Needs and Sources	PRD-1.1, PRD-1.5, PRD-2.3	FUN-3.1, FUN-6.1
VS-01.2	Establish Data Agreements and Access Rights	OVA-6, DSP-1, DSP-3	FUN-6.2, FUN-9.1, FUN-10.5
VS-01.3	Collect or Receive Data	PRD-4.1, PRD-4.2, PRD-4.3, DSP-4	FUN-3.2, FUN-9.2
VS-01.4	Validate and Log Data Intake	PRD-4.4, DLM-2	FUN-3.2, FUN-7.3
VS-01.5	Store and Register Raw Data Artefacts	DLM-3, DLM-4, DLM-5	FUN-7.2, FUN-7.3, FUN-7.5
VS-02.1	Profile and Assess Data Structure	PRD-5.3	FUN-3.3
VS-02.2	Apply Standard Classifications and Formats	PRD-5.2, DLM-9	FUN-3.3, FUN-7.4
VS-02.3	Align Metadata and Semantics	OVA-2	FUN-7.2
VS-02.4	Validate Structural and Semantic Conformance	PRD-5.3, DLM-2	FUN-3.3, FUN-7.3
VS-02.5	Register Standardized Artefacts	DLM-3, DLM-4, DLM-5	FUN-7.2, FUN-7.3, FUN-7.5
VS-03.1	Conduct Error Detection and Quality Diagnostics	PRD-5.3	FUN-3.3
VS-03.2	Perform Data Editing and Correction	PRD-5.4	FUN-3.3
VS-03.3	Impute Missing or Non-Response Data	PRD-5.4	FUN-3.3, FUN-4.1
VS-03.4	Derive Additional Variables and Indicators	PRD-5.1, PRD-5.5	FUN-3.3
VS-03.5	Perform Quality Evaluation and Versioning	PRD-5.8, DLM-2, DLM-3	FUN-3.3, FUN-5.2, FUN-7.3
VS-04.1	Design Estimation and Analytical Approach	PRD-2.4, PRD-2.5	FUN-3.1, FUN-4.1
VS-04.2	Apply Weighting and Calibration	PRD-5.6	FUN-3.4
VS-04.3	Estimate and Aggregate Data	PRD-5.7	FUN-3.4
VS-04.4	Perform Model-Based Analysis	PRD-6.1, PRD-6.3	FUN-3.4
VS-04.5	Validate and Document Results	PRD-6.2, OVA-2	FUN-3.4, FUN-7.2
VS-05.1	Select and Integrate Source Datasets	PRD-5.1, PRD-6.1	FUN-3.4
VS-05.2	Construct Composite Indicators or Indices	PRD-5.5, PRD-5.7	FUN-3.3, FUN-3.4
VS-05.3	Perform Consistency and Coherence Checks	PRD-6.2	FUN-3.4, FUN-5.2
VS-05.4	Generate Analytical and Interpretive Outputs	PRD-6.3, PRD-7.2	FUN-3.4, FUN-3.6
VS-05.5	Approve and Register Statistical Products	PRD-6.4, PRD-6.5, DLM-3	FUN-3.4, FUN-3.5, FUN-7.2

Stage	Name	Realised by processes	Main functions
VS-06.1	Plan Dissemination and Access Strategy	PRD-2.1	FUN-3.1, FUN-3.6
VS-06.2	Prepare Data for Dissemination	PRD-6.4, PRD-7.1	FUN-3.5, FUN-3.6
VS-06.3	Publish and Provide Access Services	PRD-7.2, PRD-7.3, MDS-1 to MDS-5, MDS-7, DSP-2	FUN-3.6, FUN-8.1 to FUN-8.5, FUN-9.1
VS-06.4	Support and Engage Users	PRD-7.4, PRD-7.5, MDS-6	FUN-3.6, FUN-8.3, FUN-10.9
VS-06.5	Monitor Usage and Feedback	PRD-8.1, MDS-8, MDS-9	FUN-8.1, FUN-8.3, FUN-10.9

Supporting value streams

Stage	Name	Realised by processes	Main functions
VS-07.1	Define Governance Frameworks and Policies	DLM-1	FUN-1.2, FUN-5.1, FUN-7.1
VS-07.2	Establish Legal and Regulatory Compliance Mechanisms	DLM-4	FUN-10.5, FUN-10.8
VS-07.3	Implement Data and Metadata Governance	OVA-2, OVA-3, DLM-1, DLM-6	FUN-7.1, FUN-7.2
VS-07.4	Monitor Quality and Compliance Indicators	OVA-1, DLM-8, MDS-8, DSP-5	FUN-5.2, FUN-7.1
VS-07.5	Audit, Certify and Report Compliance	OVA-1, PRD-8.2	FUN-1.2, FUN-5.2
VS-07.6	Update and Improve Governance Mechanisms	PRD-8.3, DLM-10, DLM-11	FUN-2.1, FUN-7.1
VS-08.1	Identify Innovation Needs and Opportunities	PRD-1.5	FUN-4.2
VS-08.2	Conduct Research and Proof-of-Concepts	No reference process (NSI-specific)	FUN-4.2
VS-08.3	Evaluate Methodological Soundness and Feasibility	PRD-3.6	FUN-4.1, FUN-5.2
VS-08.4	Develop and Document New Methodologies or Tools	PRD-3.2, OVA-5	FUN-4.1, FUN-10.4
VS-08.5	Integrate Innovation into Production Environments	PRD-3.7	FUN-2.4, FUN-3.1
VS-08.6	Promote Knowledge Sharing and Capability Building	OVA-5	FUN-10.1, FUN-10.4
VS-09.1	Define Digital and Technical Requirements	PRD-2.6	FUN-2.5, FUN-3.1
VS-09.2	Design Platform and Infrastructure Architecture	PRD-2.6	FUN-2.5, FUN-10.3
VS-09.3	Provision and Configure Platforms	PRD-3.4, MDS-4	FUN-8.3, FUN-10.3
VS-09.4	Enable Integration and Orchestration Services	PRD-3.4, DSP-4	FUN-9.2, FUN-10.3

Stage	Name	Realised by processes	Main functions
VS-09.5	Ensure Performance, Security and Continuity	PRD-3.5, DLM-5	FUN-10.3, FUN-10.8
VS-09.6	Evolve and Modernise the Technical Ecosystem	No reference process (NSI-specific)	FUN-2.2, FUN-10.3
VS-10.1	Identify and Segment Stakeholders	PRD-1.2, DSP-6	FUN-1.3, FUN-10.9
VS-10.2	Collect and Analyse User Needs	PRD-1.1, PRD-1.2	FUN-3.1, FUN-10.9
VS-10.3	Define and Manage Service Portfolios	PRD-2.1	FUN-10.9
VS-10.4	Deliver Services and Manage Relationships	PRD-7.5, MDS-6, OVA-6	FUN-6.1, FUN-10.9
VS-10.5	Engage and Communicate with Stakeholders	PRD-7.4	FUN-1.4, FUN-10.6
VS-10.6	Monitor Impact and Satisfaction	PRD-8.1, PRD-8.2	FUN-5.2, FUN-10.9

PLACEHOLDER: Function templates and organisation-to-function mapping instruments to be added to the toolkit

Business Services and Business Interfaces

Business Events

Business Objects, Contracts, Representations and Products

Reference Models for the Statistical Data Value Chain

Why the SAF offers more than one model NSIs differ in size, mandate, legal context, data volumes, and technology platform. A single prescribed arrangement would fit some offices well and others poorly. The SAF therefore describes reference models: alternative, internally consistent ways of organising the value chain. Each model answers the same four questions:

1. At which points is data persisted and made available for reuse?
2. Which quality criteria must data meet at each point?
3. Who is accountable for the data at each point?
4. How are confidentiality and traceability guaranteed along the chain?

The SAF currently describes two reference models. Both follow the same underlying idea, progressive improvement of data through defined layers, but they differ in granularity and in where they place governance.

Model A: Steady-State Architecture

The Steady-State Architecture organises the value chain into business domains that are loosely coupled through five numbered steady states (#1 Raw, #2 Standardized, #3 Processed, #4 Statistics, #5 Released). Each state represents a specific kind of statistical value and has its own quality gate, ownership, and handover interface. The model is derived from production practice in NSIs with many statistical domains and extensive microdata reuse. It is described in full in the remainder of this chapter.

Model B: Medallion Architecture

The Medallion Architecture organises data into three layers of increasing quality, usually called Bronze (raw), Silver (cleansed and conformed), and Gold (curated for consumption). The pattern originates from

lakehouse platforms and is a logical design pattern rather than a product. Its strength lays in its simplicity. There are three layers, a clear direction of flow, and the ability to rebuild any layer from the one below it.

As a data-engineering pattern, Medallion does not by itself define statistical semantics, accountability, or confidentiality rules per layer.

Comparison

Characteristic	Steady-State Architecture	Medallion Architecture
Number of governed handover points	Five (#1 to #5)	Three (Bronze, Silver, Gold)
Origin	Statistical production practice	Lakehouse data engineering
Granularity of quality gates	Fine: one gate per kind of statistical value	Coarse: several kinds of value combined per layer
Separation of standardization and editing	Explicit (#2 vs. #3)	Combined in Silver
Separation of internal statistics and released data	Explicit (#4 vs. #5)	Combined in Gold unless subdivided
Governance overhead	Higher: more states to own, register, and audit	Lower
Fit with platform tooling	Platform-neutral; mapping required	Native in lakehouse platforms
Typical adopters	Larger NSIs, many domains, high microdata reuse	Smaller NSIs, platform-led modernisation

Selection criteria

The criteria below help an NSI decide which model fits best. They are indicative. No single criterion is decisive, and the SAF toolkit will provide a structured assessment.

Criterion	Points towards Steady-State	Points towards Medallion
Organisational size and number of statistical domains	Many domains, many teams handing over data	Few domains, small central data team
Reuse of intermediate microdata across domains	High; standardized and processed microdata reused by several products	Limited; most data flows within one product chain
Legal requirements on pseudonymisation timing	Law or policy requires a separate, auditable pseudonymisation point	Pseudonymisation can be applied at the Silver boundary
Archival and reproducibility obligations	Specific states (e.g., processed microdata) must be retained long term	Retention can be defined per layer
Separation of internal statistics from released data	Required (embargo, different confidentiality thresholds per use)	Achievable by subdividing Gold
Metadata and governance maturity	Mature data ownership and catalogue practices in place	Governance capacity is limited and must stay lean
Technology platform	Heterogeneous landscape, on-premises and private cloud	Single lakehouse platform
Participation in international data exchange	Many handover points to external partners	Exchange concentrated on final outputs

Practice within the SAF community indicates that smaller NSIs often adopt a Medallion arrangement, while larger NSIs with many domains tend towards finer-grained states. This observation is to be substantiated through the SAF toolkit assessments.

Combining the models

The models are not mutually exclusive. The steady states can be mapped onto Medallion layers, which allows an NSI to use Medallion as its platform convention while governing data through steady states, or to start with Medallion and introduce finer states as it grows.

Steady state	Medallion layer	Note
#1 Raw Data	Bronze	Direct correspondence
#2 Standardized Data	Silver (lower sub-layer)	In Medallion often merged with #3
#3 Processed Data	Silver	Main reusable microdata layer
#4 Statistics	Gold (internal sub-layer)	Not yet released
#5 Released Data	Gold (published sub-layer)	Released after final disclosure control
#6 Data Products	Gold (consumption product sub-layer)	Packaged for consumption

Steady-State Architecture

Overview

Business domains

The statistical process is divided into three main business domains (Observation, Processing/Analysis, and Dissemination). Processing/Analysis is divided into four subdomains, which results in six business domains. Each domain adds one specific kind of statistical value and produces one steady state.

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
1	Observation/Collection	Data in NSI custody, authentic and traceable	Receive Data, Extract Data, Collect Survey	VS-01	4	#0 → #1
2	Standardization without loss of content	Common structure, format, and classifications	Transform Data, Anonymize Data, Classify Data	VS-02	5.1–5.2	#1 → #2
3	Editing, derivation, and non-response correction	Reliable, complete, enriched microdata	Imputate Data, Integrate Data	VS-03	5.3–5.5, 5.8	#2 → #3
4	Estimation and balancing	Representative statistics with accuracy	Estimate Data, Aggregate Data	VS-04	5.6–5.7, 6.1–6.4	#3 → #4

#	Business domain	Value added	Business functions (ArchiMate view)	Value stream	GSBPM	State in → out
5	Statistically composed and composite products	Coherent, approved products	Compose Data	VS-05	6.5, 7.1–7.2	#4 → #4 (approved)
6	Dissemination	Disclosure-safe, accessible released data	Data Delivery, Data Publication	VS-06	6.4, 7.3–7.5	#4 → #5
—	Consumption (external)	Use of statistics by society	Data Consumption, Consumption Optimization	VS-10 (feedback)	8	#5 → (#6)

The execution processes between the business domains are loosely coupled. Each has its own design and implementation. For the mutual exchange of data, they are loosely linked via steady state interfaces. To this end, five steady state phases are recognized as handover points at the business-domain boundaries described above. This means that the business domain "Observation" produces "Raw Data", business domain "Standardization without loss of content" produces "standardized data" etc. 1. raw data, 2. standardized data, 3. processed data, 4. statistics, 5. released data, and 6. Consumable data products.

A steady state phase is therefore an environment where high-quality versions of data are brought together for exchange, but it also represents a value in the value chain of this data.

The four process types

Each business domain uses the same four process types:

1. **Design** of the execution processes, which provides metadata.
2. **Implementation** of the execution processes, which provides the execution process.
3. **Execution** of the execution processes, which provides the statistical data.
4. **Management** of the execution processes, which provides quality reports.

How an execution process adds statistical value, and therefore how it is designed and implemented, falls outside the scope of the SAF business architecture. That design follows from, among other things, the NSI's methodology.

Design rules of the steady-state model

The model rests on six design rules. They apply the SAF architecture principles to the data value chain.

1. **Loose coupling through states (SAF-P07).** Domains exchange data only through registered steady states, never through each other's internal working data. A domain can change its methods or systems without affecting its neighbours, as long as the state it produces meets its quality gate.
2. **One state, one kind of value.** Each state represents exactly one kind of added statistical value. This keeps quality gates testable and accountability unambiguous.
3. **Immutable quality versions.** Data registered in a state is never overwritten. Corrections and redeliveries create a new version, so every statistic can be traced to the exact data it was built on.
4. **No state without metadata (SAF-P09).** Registration requires the minimum metadata set of the state. Data without it does not enter the state.

5. **Progressive confidentiality (SAF-P11).** Protection increases along the chain: direct identifiers are removed or pseudonymised at #2, disclosure risk is assessed at #3, initial disclosure control is applied at #4, and final disclosure control gates #5.
6. **Reuse before reprocessing (SAF-P08).** An authorised consumer that needs data at a given level of value reuses the registered state rather than repeating the upstream processing.

Steady states at a glance

Position	Name	Produced by	Kind of value	Mandatory?
(#0)	Source Data	External provider	Outside NSI custody, boundary position	n/a
#1	Raw Data	VS-01	Authentic, traceable custody	When the product collects its own data
#2	Standardized Data	VS-02	Comparability, values unchanged	Optional
#3	Processed Data	VS-03	Reliability and completeness of microdata	Always
#4	Statistics	VS-04, VS-05	Representative, approved statistics	Always
#5	Released Data	VS-06	Disclosure-safe, accessible output	When output is released
(#6)	Consumed Data	External users	Outside NSI control, boundary position	n/a

State lifecycle

Every steady state follows the same lifecycle:

1. **Promotion.** The producing domain submits a dataset to the quality gate of the target state.
2. **Gate outcome.** The dataset is accepted, conditionally accepted with explicit quality flags, or rejected. Rejected data does not enter the state; it returns to the producing domain or, for #1, to the provider.
3. **Registration.** An accepted dataset receives a persistent identifier, its metadata is registered in the catalogue, and a registration event notifies consuming domains.
4. **Consumption.** Authorised consuming domains read the registered version. Responsibility for further processing transfers on receipt of the event; custody of the registered version stays with the producer.
5. **Revision.** A correction produces a new version that passes the same gate. Consumers are notified and decide whether to reprocess. Earlier versions remain available for as long as statistics depend on them.
6. **Retirement.** A version is disposed of according to the retention policy of the state, after its lineage obligations have been met.

How each state is described

Each state below is described with the same ten attributes. NSIs can use this template to document their own implementation; the SAF toolkit will provide it as a fill-in form.

#	Attribute	Question it answers
1	Definition and value added	What is this state and why does it exist?

#	Attribute	Question it answers
2	Producing and consuming domains	Who creates it, and who may use it?
3	Quality gate	What must data satisfy to be registered?
4	Data and metadata characteristics	What does the data look like, and which metadata must accompany it?
5	Security, privacy, and access	How is the data protected, and who may see it?
6	Ownership and governance	Who is accountable?
7	Versioning, retention, and lineage	How are versions managed, how long are they kept, and how is the trace preserved?
8	Reuse and handover	How is the state made available to consumers?
9	Traceability to the framework	Which capabilities, value-stream stages, and GSBPM sub-processes relate to it?
10	Examples and anti-patterns	What does good practice look like, and what should be avoided?

Boundary Position #0: Source Data

Description: Source data is data as it exists at the data holder, before it enters the NSI: a register at a government agency, a company's accounting system, a respondent's answers before submission, or a web page. The NSI has no custody of source data and cannot guarantee its quality. It can influence it only through agreements, questionnaire design, and cooperation with the data holder.

Why it is described: The SAF describes #0 so that the boundary of the NSIs responsibility is explicit. Everything the NSI knows about source data is captured as metadata when the data crosses into #1: who provided it, under which agreement, through which channel, and in what condition.

State #1: Raw Data

1. Definition and value added

Raw data is data received from an external data holder or respondent, placed under NSI custody in its original form, and verified for technical integrity and authenticity. It is the first point of possession. At this moment the NSI does not have access to the data at the source, so #1 is the earliest point at which the NSI can guarantee what it holds.

The value of #1 is **authentic custody**. It proves that the NSI holds exactly what the provider delivered, when it was delivered, under which legal basis, and through which channel. That proof is the foundation for all downstream traceability. If a statistic is ever questioned, the chain of evidence ends here.

2. Producing and consuming domains

Role	Domain	Note
Producer	Observation/Collection (VS-01)	Functions: Receive Data, Extract Data, Collect Survey
Primary consumer	Standardization (VS-02)	Triggered by the *Raw Data Registered* event
Secondary consumer	Editing (VS-03)	Only where #2 is skipped for this product

Role	Domain	Note
Occasional consumer	Innovation and Methodology (VS-08)	Source exploration, under specific authorisation
Assurance	Governance and Compliance (VS-07)	Audit of provenance and legal basis

3. Quality gate

The gate for #1 checks technical and legal conditions only. It does not assess content; content quality is the concern of #3.

Criterion	Check	Typical outcome on failure
Legal basis	A valid agreement or statutory basis covers this delivery	Reject; escalate to legal
Sender authenticity	Sender verified (certificate, credentials, signed delivery)	Reject
Integrity	Checksum or digital signature matches	Reject; request redelivery
Security	Malware and security scan passed	Quarantine; reject
Structural conformance	Delivery matches the agreed technical specification (file format, schema, encoding)	Reject or conditionally accept with flag
Completeness against agreement	Record counts and delivery scope within agreed tolerance	Conditionally accept with flag, or reject
Timeliness	Delivered within the agreed window	Accept with flag; report to relationship management
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

Conditional acceptance is a deliberate option. A late or slightly incomplete delivery may still be the best available data. The flag ensures every downstream user knows this.

4. Data and metadata characteristics

Data. Raw data is stored in its original format, or in a lossless technical equivalent. Permitted technical operations are those that change representation without changing content and that can be reversed: decryption, decompression, character-encoding normalisation, and container conversion. Operations that change values, codes, or record selection are not permitted at #1. The original byte stream, or a verifiable hash of it, is retained.

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this delivery version
Provider	Organisation or respondent group that supplied the data
Agreement reference and legal basis	Agreement identifier and the applicable legal ground
Channel and method	Transfer channel (file transfer, API, data space connector, survey mode) and instrument version
Receipt timestamp	Moment of receipt into NSI custody
Integrity proof	Checksum or signature value and algorithm
Volume	Record count and size
Reference period	Period the data describes, as stated by the provider
Structural description	Schema or technical specification the delivery conforms to

Metadata element	Description
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #1 typically instantiates a *Data Set* received from an *Information Provider* through an *Exchange Channel* under a *Provision Agreement*.

5. Security, privacy, and access

Raw data usually contains direct identifiers and therefore carries the highest confidentiality rating in the chain. Access is limited to the roles that operate the intake and the standardization process. Analysts and subject-matter staff do not work on #1.

Pseudonymisation is normally applied at the transition to #2. Some NSIs pseudonymise at the intake boundary, or receive data already pseudonymised by a trusted third party. Both variants are compatible with the model, provided the choice is documented and the key management is separated from the data. Data minimisation starts before #1: the agreement and the collection design determine that only necessary data is received.

6. Ownership and governance

Role	Responsibility for #1
Data owner	Process owner of Observation/Collection; accountable for lawful acquisition and fitness of the intake
Data steward	Maintains metadata and catalogue entries; resolves gate exceptions
Data custodian	Operates storage, access control, and retention
Relationship manager	Handles provider follow-up on rejected or flagged deliveries (with VS-10)

7. Versioning, retention, and lineage

Every delivery, including a redelivery or correction by the provider, is a new immutable version. A redelivery never overwrites an earlier version, because statistics may already be built on it.

Retention of #1 is governed by two needs that pull in opposite directions. Reproducibility and audit favour keeping raw data; data minimisation and storage cost favour disposal. A common arrangement is to keep #1 until the dependent #3 version is verified and the audit period has passed, while #3 is the state subject to long-term archival requirements. The exact period is set by national law and NSI policy.

Lineage at #1 is the provenance record: the link from the registered version to the provider, the agreement, and the delivery event. Every downstream state must be able to trace back to a #1 version or, where a product starts from another domain's data, to the state it consumed.

8. Reuse and handover

Registration of #1 publishes the *Raw Data Registered* event and a catalogue entry. Consumers discover the delivery through the catalogue and access it through the handover interface, never through direct access to the intake system.

One delivery can serve several statistical products. The register-based census and the labour market statistics may both draw on the same population register delivery. Under SAF-P08, products reuse the

registered delivery rather than requesting the same data again from the provider. This also reduces response burden.

Handover element	Content
Artefact	Registered Raw Data version
Required metadata	Minimum metadata set (attribute 4)
Trigger event	*Raw Data Registered*
Responsibility transfer	VS-01 keeps custody; the consumer takes responsibility for further processing on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.2.1 Primary data collection, 1.2.2 Secondary data collection, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 3.3.4 Logical identity and access control, 3.4.3 Service agreement
Value-stream stages	VS-01.3 Collect, Receive, or Extract Data; VS-01.4 Validate and Log Data Intake; VS-01.5 Store and Register Raw Data Artefacts
GSBPM	4.3 Run collection, 4.4 Finalise collection
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11

10. Examples and anti-patterns

Examples

- *Survey.* Responses from a web questionnaire are received with paradata (mode, completion time, instrument version). The delivery is registered as one #1 version per collection batch.
- *Administrative register.* A quarterly tax file arrives by file transfer, with a manifest. Checksum, schema, and record count are checked against the agreement; 14 unknown business identifiers are not a #1 issue and pass through for handling downstream.
- *Web data.* Daily scraped price data is registered with the scraper version and page fingerprint per retailer, so a change in a website's structure can be traced to the affected deliveries.

Anti-patterns

- Correcting obvious errors "while the file is open" at intake. Any value change belongs in #3, where it is documented.
- Overwriting a delivery with the provider's corrected version. Statistics built on the first version lose their trace.
- Giving analysts direct access to raw data for convenience. This bypasses the pseudonymisation at #2.
- Keeping only a converted copy and discarding the original. The proof of authentic custody is lost.
- Registering a delivery without its agreement reference. The legal basis can no longer be demonstrated in an audit.

State #2: Standardized Data

1. Definition and value added

Standardized data is raw data that has been brought into the common structure, format, and classifications of the NSI. Its values are unchanged. Only the representation changes: fields are renamed and restructured, types are cast, source codes are mapped one-to-one to standard codes, and descriptive metadata is attached.

Direct identifiers are removed or replaced with pseudonymous keys. #2 is therefore the first state in which data from different sources can be compared and combined, and the first state without direct identifiers.

The value of #2 is **comparability without loss of content**. Data that looked different at the source now uses the same structure, the same codes, and documented definitions, units, and reference periods. Because no value has changed, the trace back to the authentic delivery in #1 stays simple and deterministic. Every interpretation, correction, or conversion is left to #3, where it is documented as an edit.

#2 is an optional state. A product can go directly from #1 to #3, for example when a source already arrives in the NSI's standard structure and coding. In that case the NSI records in the catalog that #2 was skipped, and why (see *Applying the steady states pragmatically*). **[SS1]**

2. Producing and consuming domains

Role	Domain	Note
Producer	Standardization without loss of content (VS-02)	Functions: Transform Data, Anonymize Data, Classify Data
Primary consumer	Editing (VS-03)	Triggered by the *Standardized Data Registered* event
Secondary consumer	Other statistical products	Reuse of the same standardized source under SAF-P08, with authorization
Occasional consumer	Innovation and Methodology (VS-08)	Exploration of sources and methods, under specific authorization
Assurance	Governance and Compliance (VS-07)	Audit of mapping lineage and pseudonymization

3. Quality gate

The gate for #2 checks conformance and preservation. It does not assess whether values are correct; that is the concern of #3. A value that cannot be mapped is not an error of standardization. It is a fact about the source, so it is flagged and handed to VS-03.

Criterion	Check	Typical outcome on failure
Structural conformance	Fields, types, formats, and cardinalities match the target data model	Reject; re-map
Classification conformance	Every mapped code exists in the stated classification version; no orphan codes	Reject; correct the correspondence table
Mapping coverage	Share of source codes mapped one-to-one is above the agreed threshold	Conditionally accept with flag; unmapped values listed for VS-03
Semantic metadata	Unit, currency, reference-period basis, and definition recorded for every variable	Hold until complete
No loss of content	Record counts and key measures identical to the #1 version; no value changed	Reject
Identifier protection	Direct identifiers removed or pseudonymized according to the privacy design	Reject; escalate to the privacy officer
Referential integrity	Linkage keys exist in the reference register (e.g., the statistical business register)	Conditionally accept with flag

Criterion	Check	Typical outcome on failure
Metadata and lineage	Minimum metadata set captured (see attribute 4); field-level lineage complete	Hold until complete

Conditional acceptance is the normal outcome for many sources. Unmapped codes and unmatched keys are often caused by known lags, such as newly registered businesses. The flag ensures every downstream user knows exactly which values still need work.

4. Data and metadata characteristics

Data. Standardized data is stored in the canonical target data model. Standard codes are stored next to the original source values, so the original remains visible. Permitted operations are renaming, restructuring, type casting, reformatting (for example from CSV to a standard exchange format), key alignment, one-to-one code mapping through maintained correspondence tables, removal or pseudonymization of direct identifiers, and dropping variables that are not needed for the statistical purpose. Operations that change values are not permitted at #2: unit or currency conversion, reference-period alignment, reconciliation of differing definitions, derivation, interpretive coding of free text, editing, and imputation. These belong to #3. [**SS2**]

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this standardized version
Source reference	The #1 version or versions this dataset was produced from
Target model version	Version of the canonical data model applied
Mapping ruleset and engine version	Versions of the mapping rules and the tool that applied them
Classifications applied	Scheme and version per coded variable (e.g., NACE Rev. 2, ISCED 2011)
Mapping coverage and unmapped values	Coverage per variable and the list of values routed to VS-03
Units, currency, and reference-period basis	Recorded per variable as received, not converted
Concept links and definition differences	Link to the canonical concept; recorded differences with other sources
Pseudonymization record	Identifiers removed or pseudonymized, method, key reference (never the key itself)
Conformance assessment	Structural, classification, and semantic results; no-loss-of-content confirmation
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #2 typically instantiates a *Unit Data Set* that conforms to a *Data Structure*, whose *Represented Variables* take their values from *Statistical Classifications* and *Code Lists*.

5. Security, privacy, and access

#2 is the point where direct identifiers leave the working data. **Removal** deletes an identifier and ends the possibility to link on it. **Pseudonymization** replaces it with a key that still allows records to be linked. The privacy design decides per identifier which of the two applies, and the choice is recorded in the pseudonymization record.

Pseudonymization keys are managed by the data custodian, separately from the data and under separate access rights. Staff who work on #2 and later states never have access to the keys. Some NSIs pseudonymize

at intake, or receive data already pseudonymized by a trusted third party. In that case #2 confirms and records the protection that is already in place.

Standardized data is still detailed microdata. Indirect identifiers, such as date of birth, place of residence, or occupation, remain, and combinations of them can identify a person or business. The confidentiality rating of #2 therefore stays high. Access is limited to the roles that standardize and edit the data, and to authorized reusers with a stated purpose. The formal assessment of disclosure risk starts at #3. **[SS3]**

6. Ownership and governance

Role	Responsibility for #2
Data owner	Process owner of Standardization; accountable for conformance and for applying the privacy design
Data steward	Registers versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, and retention; manages pseudonymization keys separately from the data
Classification steward	Maintains the correspondence tables and the classification versions used in mapping
Metadata and data architect	Owns the target data model and the links to canonical concepts

7. Versioning, retention, and lineage

Every standardization run produces a new immutable version. A new version also arises when the input stays the same but the rules change: a new correspondence table, a new classification version, or a revised target model. The earlier version is never overwritten, because #3 versions may already be built on it. Each #2 version records the exact #1 version and rule versions it was produced from.

Standardization is deterministic, so a #2 version can be regenerated from its #1 version and the recorded rule versions. Many NSIs therefore keep #2 only as long as the dependent #3 versions are produced and reviewed. This only works while #1 and the rule versions are themselves retained. If #1 is disposed of earlier, #2 must be kept for as long as reproducibility requires. The exact periods are set by national law and NSI policy.

Lineage at #2 is field-level. Each standardized value can be traced to the raw value it came from and to the mapping rule that was applied. The pseudonymization record shows which identifiers were replaced, without revealing the keys.

8. Reuse and handover

Registration of #2 publishes the *Standardized Data Registered* event and a catalog entry. Consumers discover the version through the catalog and access it through the handover interface, never through the standardization working environment.

One standardized source can serve many statistical products. A population register delivery is standardized once and is then used by the census, labour market statistics, and income statistics. Under SAF-P08, products reuse the registered version instead of standardizing the same source again. Repeating the work per product wastes effort and leads to inconsistent coding of the same source.

The list of unmapped values and the recorded units and periods travel with the handover. VS-03 then knows exactly which interpretive coding and conversions remain to be done.

Handover element	Content
Artifact	Registered Standardized Data version

Handover element	Content
Required metadata	Minimum metadata set (attribute 4), including unmapped values and recorded units and periods
Trigger event	*Standardized Data Registered*
Responsibility transfer	VS-02 keeps custody; the consumer takes responsibility for editing, interpretive coding, and conversion on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.3.1 Data standardization, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.4 Statistical register management, 3.3.1 Privacy management, 3.3.4 Logical identity and access control
Value-stream stages	VS-02.1 Profile and Assess Data Structure; VS-02.2 Apply Standard Classifications and Formats; VS-02.3 Align Metadata and Semantics; VS-02.4 Validate Structural and Semantic Conformance; VS-02.5 Register Standardized Artifacts
SAF processes [**SS4**]	PRD-5.2 Classify and code (deterministic mapping only), PRD-5.3 Review and validate (structural checks only), DLM-2 Promote data through a quality gate, DLM-3 Register and catalog data, DLM-4 Classify and protect data, DLM-9 Manage reference and master data
GSBPM	5.1 Integrate data (key alignment only; record linkage takes place in #3), 5.2 Classify and code
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer [**SS4**]	Silver, lower sub-layer; in a Medallion arrangement often merged with #3

10. Examples and anti-patterns

Examples

- *Survey.* Labour Force Survey returns are restructured into the canonical person and household model. Source education codes are mapped one-to-one to ISCED 2011, and region codes to NUTS-3. The 42,000 free-text occupation answers are not coded here; they are listed as unmapped and handed to VS-03. The personal identifier is replaced with a pseudonymous key.
- *Administrative register.* The tax authority's internal activity codes are mapped to NACE Rev. 2 through a maintained correspondence table; 0.6% of codes have no correspondence and are flagged. Turnover stays in thousands of euro on a fiscal-year basis, and this basis is recorded as metadata. Conversion to units on a calendar-year basis takes place in #3.
- *Web data.* Scraped price records are reshaped into the canonical price record. Currency, unit, and scrape timestamp are recorded as metadata, and prices are kept exactly as received. Most products only have a free-text name and are passed to VS-03 for classification to COICOP.

Anti-patterns

- Converting currencies or units at #2 "because the data is open anyway". The value changes without an edit record.
- Coding free text with a classifier at #2. Interpretive coding needs review and logging, which happens in #3.
- Silently dropping records with unknown codes. This breaks the no-loss-of-content guarantee.

- Overwriting a #2 version after a correspondence table update. Statistics built on the earlier version lose their trace.
- Storing pseudonymization keys together with the data. The protection of #2 is then only on paper.
- Skipping #2 without recording the reason. Auditors can no longer see why the state is missing.

State #3: Processed Data

1. Definition and value added

Processed data is microdata whose content has been checked, corrected, completed, and enriched. Errors, duplicates, outliers, and inconsistencies have been detected and resolved. Missing values have been imputed, free text has been coded, and units, currencies, and reference periods have been converted to a common basis. Records from different sources have been linked, and new variables have been derived. Unlike in #2, values do change here, but every change is documented, attributable, and reversible.

The value of #3 is **reliability and analytical readiness**. Every estimate, model, and publication downstream rests on this microdata. The guarantee is not that values stay the same, but that no value changes without a record of the original value, the rule or decision applied, the reason, the actor, and the time. That record makes it possible to explain, reproduce, or reverse any change.

#3 is the only steady state that is mandatory for all statistical production. It is the microdata on which everything downstream is verifiably built. It is also the state that national archival requirements usually apply to: without it, the statistics built on it cannot be re-verified. **[SS5]**

Processing is primarily driven by the individual statistical domains. Depending on the type of data, it may also be performed by a central unit for efficiency and consistency. Either way, the result is registered as #3 under the same quality gate.

2. Producing and consuming domains

Role	Domain	Note
Producer	Editing, derivation, and non-response correction (VS-03)	Functions: Impute Data, Integrate Data
Primary consumer	Estimation and Analysis (VS-04)	Triggered by the *Processed Data Registered* event
Secondary consumer	Other statistical products	Reuse of processed microdata under SAF-P08, with authorization
Secondary consumer	Microdata services (VS-06)	Research datasets are prepared from #3 and released as #5 **[SS6]**
Occasional consumer	Innovation and Methodology (VS-08)	Testing of new editing, imputation, or coding methods, under specific authorization
Assurance	Governance and Compliance (VS-07)	Audit of edit lineage, methodology compliance, and disclosure risk

3. Quality gate

The gate for #3 checks content quality and traceability. It is the first gate that judges whether values are plausible and complete. A dataset may pass with known remaining quality issues, as long as they are measured and documented.

Criterion	Check	Typical outcome on failure
Rule conformance	No record violates the validation and editing rules after correction	Reject; re-edit
Edit lineage	Every changed value has its original value, rule or decision, reason, actor, and time recorded	Hold until complete
Imputation flags	Every imputed value carries a flag and a method code; no observed value is overwritten	Reject
Coding	Free-text values are coded or explicitly left uncoded with a reason; low-confidence codes are reviewed	Conditionally accept with flag
Conversion and reconciliation	Every unit, currency, or reference-period conversion and every reconciliation has its rule and original value recorded	Hold until complete
Linkage quality	Linkage rate and match quality are measured and within agreed limits	Conditionally accept with flag, or reject
Derived variables	Every derived variable has a formula, inputs, unit, and classification	Hold until complete
Quality improvement	Completeness and accuracy measured against the #2 baseline (or #1 where #2 was skipped)	Conditionally accept with flag
Disclosure risk	Re-identification risk of the enriched and linked data assessed against the privacy design	Reject; reduce detail or restrict access
Reproducibility	The version can be regenerated from its inputs plus the recorded rules and models	Reject

Conditional acceptance is common for imputation and linkage. A linkage rate slightly below target may still give the best available data. The flag makes sure estimation in VS-04 takes the issue into account, for example in its variance estimation.

4. Data and metadata characteristics

Data. Processed data contains the processed value of every variable. The original value stays recoverable through the edit lineage, and imputed values carry a flag. Permitted operations are error detection, editing and correction, imputation, interpretive coding of free text, conversion of units, currencies, and reference periods, reconciliation of values between sources, record linkage, and derivation of new variables and units. These are the value-changing operations that #2 deliberately left open. Operations that turn microdata into statistics are not permitted at #3: weighting, calibration, estimation, and aggregation belong to #4. **[SS5]**

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this quality version
Source reference	The #2 version or versions (or #1, where #2 was skipped) this dataset was produced from
Rule and model versions	Versions of the editing rules, imputation models, coding service, and derivation specifications applied
Edit lineage	Per changed value: original value, new value, rule or decision, reason code, actor, and time

Metadata element	Description
Imputation record	Per imputed value: flag, method code, and the predictors or donors used
Coding record	Per coded value: source text, assigned code, method, and confidence
Conversion and reconciliation record	Rule, factor or rate, and original value for every conversion or reconciliation
Linkage record	Keys used, matching method, linkage rate, and match quality
Derived variables	Formula, inputs, unit, and classification per derived variable
Quality assessment	Edit rate, imputation rate, coding completion, and improvement against the baseline
Disclosure-risk assessment	Outcome of the re-identification risk assessment and the resulting access conditions
Gate outcome	Accepted or conditionally accepted, with quality flags
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #3 typically instantiates a *Unit Data Set*, extended with derived *Represented Variables*. The edits are described as *Rules* applied in *Process Steps*, which gives the edit lineage its formal structure.

5. Security, privacy, and access

Processed data is pseudonymized microdata, but it is richer than #2. Linking sources and deriving variables adds detail, and more detail means a higher risk that a person or business can be recognized. #3 is therefore the point where the disclosure risk of microdata is formally assessed. The outcome sets the conditions for access and reuse.

Linkage uses the pseudonymous keys created at #2. The keys themselves remain with the data custodian, separately from the data. Access to #3 is limited to the roles that edit, estimate, and analyze the data, and to authorized reusers with a stated purpose. External researchers never access #3 directly. Their access runs through the microdata services in VS-06, which prepare a project-specific dataset with its own pseudonyms and release it as #5.

The edit audit trail is itself sensitive and must be tamper-evident. It shows who changed which value and why, which is essential evidence in an audit.

6. Ownership and governance

Role	Responsibility for #3
Data owner	Process owner of Editing; accountable for the quality version and for the correct application of approved methods
Data steward	Registers quality versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, retention, and archival; keeps pseudonymization keys separate from the data
Methodologist	Owns the editing rules, imputation methodology, and derivation specifications
Subject-matter statistician	Adjudicates flagged records and signs off domain-specific corrections

7. Versioning, retention, and lineage

Every completed editing cycle produces a new immutable quality version. A new version also arises when a source is redelivered, when a correction comes in, or when a rule or model changes. The earlier version is never overwritten, because estimates may already be built on it. Many NSIs distinguish preliminary and final versions; both are registered as separate quality versions of #3.

#3 is usually the state with the longest retention. It is the microdata that must be preserved to re-verify published statistics, and archival legislation often applies to it. When #3 is archived, its edit lineage, imputation flags, and derived-variable definitions are archived with it. Without that metadata, the archived values cannot be interpreted.

Lineage at #3 is value-level. Each processed value can be traced back to its original value in #2 or #1, the rule or decision that changed it, and the inputs of any imputation, conversion, or derivation. Linked records keep a reference to the source records they were built from.

8. Reuse and handover

Registration of #3 publishes the *Processed Data Registered* event and a catalog entry. Consumers discover the version through the catalog and access it through the handover interface, never through the editing working environment.

Processed microdata is the most valuable reusable asset in the chain. An integrated person-level dataset, edited and enriched once, can serve many statistics on labor, income, education, and health. Under SAF-P08, products reuse the registered quality version rather than editing the same data again. Editing the same data twice leads to different corrections of the same record, and therefore to statistics that do not agree.

The imputation flags travel with the handover. VS-04 needs them to reflect the extra uncertainty of imputed values in its accuracy measures.

Handover element	Content
Artifact	Registered Processed Data quality version
Required metadata	Minimum metadata set (attribute 4), including edit lineage, imputation flags, and derived-variable definitions
Trigger event	*Processed Data Registered*
Responsibility transfer	VS-03 keeps custody; the consumer takes responsibility for weighting, estimation, and analysis on receipt of the event

9. Traceability to the framework

Framework element	References
Capabilities	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing, 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management
Value-stream stages	VS-03.1 Conduct Error Detection and Quality Diagnostics; VS-03.2 Perform Data Editing and Correction; VS-03.3 Impute Missing or Non-Response Data; VS-03.4 Derive Additional Variables and Indicators; VS-03.5 Perform Quality Evaluation and Versioning

Framework element	References
SAF processes	PRD-5.1 Integrate data, PRD-5.2 Classify and code (interpretive coding), PRD-5.3 Review and validate, PRD-5.4 Edit and impute, PRD-5.5 Derive new variables and units, PRD-5.8 Finalize data files, DLM-2, DLM-3, DLM-7 Revise data, DLM-10 Archive and preserve data
GSBPM	5.1 Integrate data, 5.2 Classify and code, 5.3 Review and validate, 5.4 Edit and impute, 5.5 Derive new variables and units, 5.8 Finalize data files
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Silver; the main reusable microdata layer

10. Examples and anti-patterns

Examples

- *Survey.* The 42,000 free-text occupation answers handed over from #2 are coded to ISCO-08: 88% automatically, the rest computer-assisted or by experts. Item non-response on income (9%) is imputed by regression and flagged. Each coded and imputed value carries its method, so it can be reviewed or revised later.
- *Administrative register.* Turnover, received in thousands of euro on a fiscal-year basis, is converted to units on a calendar-year basis. The conversion factor and the original value are stored in the lineage. The tax data is then linked to the business survey on the statistical business register key, and a reconciliation rule resolves values where the two sources disagree.
- *Web data.* Products with only a free-text name are classified to COICOP with a machine-learning classifier; low-confidence cases are reviewed. Prices are converted to euro and standard units. Missing daily prices are filled by carry-forward and flagged with a staleness indicator.

Anti-patterns

- Overwriting the original value with the corrected one. The edit can no longer be explained or reversed.
- Imputing without a flag. Estimation then treats imputed values as observed and understates uncertainty.
- Correcting every outlier by hand. Over-editing costs effort, changes little in the results, and can introduce bias.
- Calculating weights or aggregates at #3. That work belongs to #4, where its method and accuracy are documented.
- Fixing values in a spreadsheet outside the editing pipeline. The change has no lineage and cannot be reproduced.
- Linking an extra source without reassessing disclosure risk. The richer dataset may identify people that #2 did not.
- Disposing of #3 once figures are published. The published statistics can then no longer be verified.

State #4: Statistics

1. Definition and value added

Statistics are the results produced from processed microdata: estimates, aggregates, indices, time series, and model outputs. Each result carries the method that produced it and a measure of its accuracy, such as a standard error or confidence interval. State #4 can also hold output microdata with weights, prepared for a specific internal purpose. Initial disclosure control has been applied, so the results can potentially be released. They are not yet released: #4 may still contain detail that is confidential or not suitable for publication.

The value of #4 is **statistical credibility and coherence**. Data has become information about a population, and that information is accurate, transparent, and consistent with related statistics. Any figure

in #4 can be regenerated from #3 and defended, because its method, parameters, weights, and accuracy are recorded.

#4 is produced by two domains and therefore has **two gate moments**. [**SS7**]

Gate moment	Produced by	Status of the version	What it confirms
Analytical gate	Estimation and closed-loop (VS-04)	*Analytical*	Results are methodologically sound, accurate, coherent, and initially disclosure-controlled
Approval gate	Statistically composed and composite products (VS-05)	*Approved*	Results are composed into a coherent product, explained, checked against dissemination and confidentiality standards, and signed off

Both are versions of the same state. Approval does not change an analytical version; it creates a new, immutable version with the status *approved* that refers to the analytical versions it was composed from. #4 is mandatory for all statistical production.

2. Producing and consuming domains

Role	Domain	Note
Producer (analytical)	Estimation and closed-loop (VS-04)	Functions: Estimate Data, Aggregate Data
Producer (approved)	Statistically composed and composite products (VS-05)	Function: Compose Data
Primary consumer (analytical)	Product composition (VS-05)	Triggered by the *Output Data Registered* event
Primary consumer (approved)	Dissemination (VS-06)	Triggered by the *Statistical Product Registered* event
Secondary consumer	Other statistical processes	Internal input for macro statistics and integration systems, such as the national accounts
Assurance	Governance and Compliance (VS-07)	Audit of methodology, reproducibility, and approval

3. Quality gate

The two gate moments check different things. The analytical gate tests the statistics; the approval gate tests the product.

Analytical gate (VS-04)

Criterion	Check	Typical outcome on failure
Approved method	Estimation, weighting, and model methods are approved and documented for this output	Reject; return to design
Weights	Calibration converged; weights within the agreed bounds	Reject; re-weight

Criterion	Check	Typical outcome on failure
Accuracy	Every estimate has an accuracy measure; imputation uncertainty from #3 is reflected	Hold until complete
Model diagnostics	Diagnostics of model-based results are within the agreed limits	Reject, or conditionally accept with flag
Soundness and coherence	Results are plausible and coherent with prior periods and related statistics; large movements are explained	Conditionally accept with flag, or reject
Initial disclosure control	Disclosure risk assessed and initial protection applied	Reject
Reproducibility	Results regenerate from #3 plus the recorded methodology and parameters	Reject
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

Approval gate (VS-05)

Criterion	Check	Typical outcome on failure
Component selection	The exact analytical versions used are pinned and recorded	Hold until complete
Composition	Composite indicators, indices, and time series follow their documented specifications	Reject; re-compose
Consistency and coherence	Totals reconcile to components; the product is coherent with related and prior products	Return to VS-04, or conditionally accept with explained flag
Interpretive content	Commentary and visualizations are reviewed and consistent with the data and its accuracy	Hold until complete
Dissemination and confidentiality standards	The product meets the standards, including the disclosure risk of combining outputs	Reject
Sign-off	The designated approval authority has approved the product	Hold until approved

Coherence problems at the approval gate are never solved by adjusting a component estimate in VS-05. If a component is wrong, it goes back to VS-04 and returns as a new analytical version.

4. Data and metadata characteristics

Data. Statistics in #4 are mostly aggregated data: tables, data cubes, indicators, indices, and time series. They can also include weighted output microdata, prepared for a specific internal use. Permitted operations are weighting and calibration, estimation and aggregation, model-based analysis, seasonal adjustment, initial disclosure control, and, at the approval stage, composition of indicators, indices, and time series from analytical versions. Changing microdata values is not permitted at #4: corrections go back to #3. Final disclosure control for release is not part of #4 either; it belongs to the transition to #5. **[SS8]**

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this version
Status	*Analytical* or *approved*
Source reference	The #3 versions (for analytical versions) or the analytical #4 versions (for approved versions) used
Methodology	Estimation, weighting, and model methods, with their versions and parameters
Weights and calibration totals	Calibration method, auxiliary variables, and the source and version of the totals
Accuracy	Standard error, coefficient of variation, or confidence interval per estimate; whether imputation uncertainty is included
Model diagnostics	Fit and prediction intervals for model-based results
Disclosure control	Method applied and outcome of the risk assessment
Coherence results	Outcome of the coherence checks against related and prior statistics
Composition record	For approved versions: composition method and the component versions used
Interpretive content	For approved versions: reference to commentary, key messages, and visualizations
Approval record	For approved versions: approval authority, date, and confidentiality clearance
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, aggregated results in #4 typically instantiate a *Dimensional Data Set*, and weighted output microdata a *Unit Data Set*. Both conform to a *Data Structure* that SDMX can use for exchange.

5. Security, privacy, and access

Initial disclosure control is applied at the analytical gate, so the results could in principle be released. That does not make #4 public. It may still hold detailed cells, internal breakdowns, or output microdata that are only suitable for internal use. Different confidentiality thresholds can apply depending on the intended use, and the version records which threshold was applied.

Composition creates a specific risk. Two tables that are each safe on their own can together reveal a value through differencing. The approval gate therefore checks the disclosure risk of the product as a whole, not only of its components.

Approved products are usually market-sensitive or politically sensitive before their release. Access to approved versions is restricted to the people who need it to prepare the release. Every pre-release access is logged, so equal access at the moment of release can be demonstrated.

6. Ownership and governance

Role	Responsibility for #4
Data owner (analytical)	Process owner of Estimation; accountable for methodological soundness and accuracy
Data owner (approved)	Process owner of Product Composition; accountable for coherence and interpretability of the product
Approval authority	Signs off approved versions against dissemination and confidentiality standards
Data steward	Registers versions, maintains catalog entries, and resolves gate exceptions
Data custodian	Operates storage, access control, retention, and the pre-release access log
Methodologist	Owns estimation, weighting, and model methods

Role	Responsibility for #4
Disclosure control specialist	Owns the disclosure-control rules for internal and potential external use

7. Versioning, retention, and lineage

Every estimation run produces a new immutable analytical version. Every approval produces a new immutable approved version that refers to the analytical versions it was built from. Revisions follow the revision policy: a preliminary figure, a revised figure, and a final figure are three separate versions. Nothing is overwritten, because published figures and other statistics may depend on any of them.

Approved versions that were released are kept for as long as the released figures exist, because they are the direct basis of #5. Analytical versions that never led to a release can be kept for a shorter period, as long as revisions and reproducibility are covered. The exact periods are set by national law and NSI policy.

Lineage at #4 has two steps. An estimate traces back to the #3 version, the method, the parameters, and the weights that produced it. An approved product traces back to the analytical versions it was composed from, and through them to #3.

8. Reuse and handover

#4 has two handovers, one for each gate moment. The analytical version goes to VS-05 for composition; the approved version goes to VS-06 for release. Consumers access versions through the catalog and the handover interface, never through the estimation or composition working environment.

Statistics in #4 are also reused internally. Integration systems, such as the national accounts or labor accounts, use statistics from many domains as input. Under SAF-P08 they reuse the registered version instead of estimating the same figure again. This keeps internal use and later publication consistent.

Handover element	Analytical version (VS-04 → VS-05)	Approved version (VS-05 → VS-06)
Artifact	Registered analytical #4 version	Registered approved #4 version
Required metadata	Methodology, weights, accuracy, model diagnostics, disclosure control, coherence results	Composition record, propagated accuracy, coherence results, interpretive content, approval record
Trigger event	*Output Data Registered*	*Statistical Product Registered*
Responsibility transfer	VS-04 keeps custody; VS-05 takes responsibility for composition and approval	VS-05 keeps custody; VS-06 takes responsibility for final disclosure control, release, and user support

9. Traceability to the framework

Framework element	References
Capabilities	1.3.5 Estimating population parameters, 1.3.6 Integrate statistics, 1.3.7 Statistical disclosure, 1.4.1 Statistical (output) analysis, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Value-stream stages	VS-04.1 to VS-04.5 (analytical gate); VS-05.1 to VS-05.5 (approval gate)

Framework element	References
SAF processes	PRD-5.6 Calculate weights, PRD-5.7 Calculate aggregates, PRD-6.1 Prepare draft outputs, PRD-6.2 Validate outputs, PRD-6.3 Interpret and explain outputs, PRD-6.4 Apply disclosure control (initial), PRD-6.5 Finalize outputs, PRD-7.2 Produce dissemination products (content), DLM-2, DLM-3, DLM-7 Revise data
GSBPM	5.6 Calculate weights, 5.7 Calculate aggregates, 6.1 to 6.5 Analyze
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Gold, internal sub-layer; not yet released

10. Examples and anti-patterns

Examples

- *Survey.* The Labor Force Survey estimates an employment rate of 71.4%, with a 95% confidence interval of ±0.5 percentage points at national level. The variance includes the uncertainty of imputed income from #3. This analytical version is then composed with unemployment and vacancy figures into the quarterly labor market product, which is approved and registered.
- *Administrative register.* Turnover is aggregated by industry and region. For small domains where the direct aggregate is unstable, a small-area model is used, and its results are flagged as model-based. In the approval gate, one domain fails the coherence check; it goes back to VS-04 instead of being adjusted in the product.
- *Web data.* A multilateral price index is calculated from the item prices in #3. The headline index and its main divisions are composed with the period weights, checked for coherence with the prior month and year, and approved.

Anti-patterns

- Approving a product by overwriting the analytical version. The difference between the tested figure and the approved figure disappears.
- Adjusting a component estimate in VS-05 to make the coherence check pass. The estimate no longer matches its documented method.
- Publishing an estimate without an accuracy measure. Users cannot judge whether a movement is meaningful.
- Treating initial disclosure control as final. Release requires the final check of the transition to #5.
- Sharing approved figures before release without logging the access. Equal access can no longer be demonstrated.
- Re-estimating a figure for internal use instead of reusing the registered version. Internal and published figures start to differ.

State #5: Released Data

1. Definition and value added

Released data is statistical output that the NSI has made available outside its own organization. It is based on an approved #4 version that has passed final disclosure control, has been documented for correct interpretation, and has been released through a trusted channel. Released data includes published tables and datasets, open data, data available through an API, deliveries to international organizations, data offered in a data space, and de-identified microdata made available to accredited researchers. #5 is the last point at which the NSI controls the data. Once users have it, it becomes consumed data, outside the NSI's control.

The value of #5 is **accessibility and trust**. Users get accurate, documented statistics through channels they can rely on, at a moment that is the same for everyone. The final disclosure check ensures that no

respondent can be identified. The documentation ensures that users can interpret the figures correctly, including their accuracy and limitations.

#5 is mandatory whenever output is released. If nothing changes between the approved #4 version and the release, the same artifact can be marked as released instead of being registered again. The release steps still take place and are recorded: release authorization, the final disclosure check, and the release itself (see *Applying the steady states pragmatically*).

2. Producing and consuming domains

Role	Domain	Note
Producer	Dissemination (VS-06)	Functions: Data Delivery, Data Publication
Primary consumer	External users	Policy makers, researchers, media, businesses, and the public; use turns #5 into #6
Primary consumer	International organizations	Deliveries under agreed transmission programs, such as to Eurostat or the OECD
Primary consumer	Accredited researchers	Access to de-identified research datasets in a secure processing environment **[SS9]**
Secondary consumer	Customer and Stakeholder Engagement (VS-10)	Usage and feedback data for needs analysis and impact measurement
Assurance	Governance and Compliance (VS-07)	Audit of disclosure control, equal access, and release authorization

3. Quality gate

The gate for #5 is the release gate. Confidentiality is not negotiable at this boundary. Unlike the earlier gates, the disclosure criteria allow no conditional acceptance: a release that fails them is returned for further protection. **[SS10]**

Criterion	Check	Typical outcome on failure
Approved source	The release is based on an approved #4 version	Reject
Final disclosure control	Final protection applied and disclosure-risk criteria met for every table, dataset, and dashboard	Reject; apply further protection
De-identification	Released microdata meets the de-identification standard for its channel	Reject
Documentation	Methodology notes, metadata, and user guidance are complete	Hold until complete
Format and accessibility	Each channel format is valid, machine-readable where required, and meets the accessibility standard	Hold until corrected
Release authorization	The release has been signed off	Hold until authorized
Timing and equal access	Release time follows the release calendar; embargo arrangements are in place and logged	Hold until the release slot
Metadata	Minimum metadata set captured (see attribute 4)	Hold until complete

4. Data and metadata characteristics

Data. One released version can have several representations, one per channel: a published table, an SDMX dataset behind an API, a dashboard, or a research dataset. All representations come from the same released version, so a user always gets the same figure through every channel. Permitted operations are final disclosure control, de-identification of microdata, formatting per channel, packaging, and adding documentation. Changing the statistical content is not permitted at #5. A correction to a released figure goes back to #4 as a new approved version and is released again with an erratum. [**SS11**]

Minimum metadata set.

Metadata element	Description
Persistent identifier and version	Unique identifier of this release version
Source reference	The approved #4 version this release is based on
Final disclosure control	Method applied, risk assessment outcome, and sign-off
De-identification	For microdata: the standard applied and the project or channel it applies to
Channels and formats	Every channel used, with format and access conditions
Release timestamp	Moment of release and reference to the release calendar
Embargo record	Who had pre-release access, when, and under which agreement
Documentation	References to methodology notes, reference metadata, and user guidance
Accessibility	Accessibility standard met per channel
Terms of use	Licence or terms under which the data may be used
Correction history	Errata issued and the versions they replace
Classification	Availability, integrity, and confidentiality rating

In GSIM terms, #5 is the disseminated form of a *Data Set* together with its *Referential Metadata*. SDMX defines the structure in which it is exchanged with users and international organizations.

5. Security, privacy, and access

After final disclosure control, most released data is public and has a low confidentiality rating. Its integrity rating is high. Users must be able to trust that a figure on the website or in the API is exactly the figure the NSI released. Channels therefore protect released data against tampering, and the NSI can prove which version was released when.

Before the release moment, the same data is highly restricted. Embargoed copies go only to parties with a formal agreement, such as accredited media, and every pre-release access is logged. No user gets the data before the release time outside that policy.

Research microdata follows a different path. It is released only to accredited researchers and only inside a secure processing environment. Each project gets its own dataset with project-specific pseudonyms, so data cannot be linked across projects. Results leave the environment only after an output check.

6. Ownership and governance

Role	Responsibility for #5
Data owner	Process owner of Dissemination; accountable for confidentiality, equal access, and the quality of the release

Role	Responsibility for #5
Release manager	Controls release timing, embargo, and equal access across channels
Disclosure control specialist	Applies and signs off the final disclosure control
Dissemination and service manager	Manages channels, formats, service levels, and the research access service
Data steward	Registers release versions, maintains documentation, and handles errata
Data custodian	Operates channels, access control, integrity protection, and the embargo log

7. Versioning, retention, and lineage

Every release is an immutable version. A correction never silently replaces a released figure. It produces a new approved #4 version, a new release version, and a public erratum that explains what changed and why. Earlier releases remain identifiable, so users can see which figure was available at which moment.

Released statistics are usually kept permanently, as part of the public record. National archival legislation often prescribes this. Research datasets are the exception: they are disposed of when the research project closes, together with their project-specific pseudonymization keys. The approved code and outputs of the project are kept for reproducibility.

Lineage at #5 closes the chain. Each released figure traces back to its approved #4 version, and from there through the analytical versions, #3, and #2 to the authentic delivery in #1. This makes it possible to answer any question about a published figure, from source to release.

8. Reuse and handover

Release publishes the *Statistics Released* event and makes the data available through the agreed channels. #5 has no downstream production stream. Its handover is to users outside the NSI, where the data becomes consumed data (#6). Usage and feedback flow back to VS-06 and VS-10, and through them into the design of the whole chain.

Released data is the most widely reused form of NSI output. Under SAF-P08 it is offered as openly as confidentiality allows, in standard formats and with clear terms of use. Open, documented release reduces the number of individual data requests and lets users combine NSI data with other sources.

Handover element	Content
Artifact	Released data version, in its channel-specific representations
Required metadata	Minimum metadata set (attribute 4), including documentation, terms of use, and correction history
Trigger event	*Statistics Released*
Responsibility transfer	The NSI remains responsible for the correctness and documentation of the release and for issuing errata; users are responsible for how they use the data

9. Traceability to the framework

Framework element	References
Capabilities	1.3.7 Statistical disclosure, 1.5.1 Publication management, 1.5.2 Promotion to use, 1.5.3 Facilitate data access, 1.5.4 Statistical information management, 2.2.4 Data services, 3.4.1 Services management

Framework element	References
Value-stream stages	VS-06.1 Plan Dissemination and Access Strategy; VS-06.2 Prepare Data for Dissemination; VS-06.3 Publish and Provide Access Services; VS-06.4 Support and Engage Users; VS-06.5 Monitor Usage and Feedback
SAF processes	PRD-6.4 Apply disclosure control (final), PRD-7.1 to PRD-7.5 Disseminate, MDS-3 Prepare research datasets, MDS-4 Provide a secure processing environment, MDS-7 Check and release output, MDS-9 Close projects, DSP-2 Publish data offerings, DSP-4 Exchange data under usage control
GSBPM	6.4 Apply disclosure control, 7.1 to 7.5 Disseminate
Principles	SAF-P07, SAF-P08, SAF-P09, SAF-P10, SAF-P11
Medallion layer	Gold, published sub-layer

10. Examples and anti-patterns

Examples

- *Survey.* The quarterly labor market figures are released at 09:00 on the calendar date, at the same moment on the website, the API, and the dashboard. Two small regional cells are suppressed in the final disclosure check. Accredited media received an embargoed copy and publish at the same moment.
- *Administrative register.* Aggregated business statistics are published as open data. De-identified business microdata is released to accredited researchers inside a secure processing environment, with project-specific pseudonyms and output checking.
- *Web data.* The monthly price index is market-sensitive. It is released at a fixed minute with strict equal access, and the documentation explains the main contributions to the monthly change.

Anti-patterns

- Publishing directly from #4 without a recorded final disclosure check. There is no evidence that the release was safe.
- Correcting a published figure silently. Users who already used the old figure cannot see that it changed.
- Giving a ministry or partner early access outside the embargo policy. The NSI's independence and equal access are compromised.
- Releasing figures without methodology notes or accuracy. Users draw conclusions the data cannot support.
- Sending research microdata by email or file transfer to a researcher's own environment. Control over the data is lost.
- Releasing different figures through different channels. Users lose trust when the website and the API disagree.

Boundary Position #6: Consumed Data

Description: Consumed data is released data that users have taken into their own environment and use for their own purposes. Examples are a government departments using an unemployment rate in a policy note, a researcher combining released tables with other sources, a journalist quoting an inflation figure, or an application that pulls a series from the API every day. The NSI has no custody of consumed data and cannot control how it is used, combined, or presented. It can influence correct use only through the quality of its documentation, its terms of use, its user support, and its errata.

Why it is described: The SAF describes #6 so that the end of the NSI's responsibility is explicit, just as #0 marks where it begins. Everything the NSI can do for correct use has to be done at #5: documentation, accuracy measures, terms of use, and a clear correction history. What the NSI learns about consumed data

comes back as usage and feedback, through VS-06.5 and VS-10. That feedback closes the loop: it shapes new information needs and, through them, the next cycle of the value chain from #0 onward.

The Statistical Data Value Chain

The SAF business layer describes how a National Statistical Office (NSI) turns externally-held data into trusted statistics that serve society. Six **primary** value streams form a linear production chain, each adding a specific kind of statistical value, and four **supporting** value streams enable them across the board.

> **i** Key concepts — if you are new to architecture documents
>
> This documentation uses a handful of architecture terms. Here is what they mean in plain language:
>
> - **Value stream** — a sequence of steps that together create a statistical output. Think of it as a production line: VS-01 to VS-06 each add a specific kind of value to the data.
> - **Value stream stage vs. business process** — the stage (`VS-0x.y`) says *what value* is added; the business process describes *how* your office actually does it. The numbering maps one-to-one.[1]
> - **Steady state** — a fixed handover point: a versioned, quality-assured dataset that one stream produces and the next one consumes. The framework defines five numbered steady states, #1 to #5; data still at the external provider (#0) and data in users' hands (#6) sit outside the NSI's control and are boundary positions, not numbered states.
> - **The four process types** — every domain is described through the same four groups: **Design** (decide how it should work), **Implementation** (build and test it), **Execution** (run it each production cycle), **Management** (check quality and feed improvements back).
> - **Business service / object / event** — what a domain offers to others / the data artefacts it works on / the signals streams send each other (e.g. *"Raw Data Registered"* tells the next stream new data is ready).
> - **NSI** — National Statistical Office.
> - **GSBPM** — the *Generic Statistical Business Process Model*: the standard, internationally agreed description of the phases of statistical production (specify needs, design, build, collect, process, analyse, disseminate, evaluate).
> - **GAMSO** — the *Generic Activity Model for Statistical Organizations*: the companion standard describing management and support activities (strategy, capability development, corporate support).
> - **SAF-P01 ... SAF-P11** — the eleven SAF architecture **p**rinciples, defined in the SAF Framework document and referenced throughout.

This document is the consolidated, easier-to-read view of all ten streams. The shared concepts the primary streams rely on — the chain itself, the steady states of data, the four-process governance pattern, and the principles — are explained once below; each **primary** stream is then summarized compactly, keeping its execution detail. The four **supporting** streams follow at the end, in a section that first explains how a supporting stream differs from a production stream. For the full design, implementation, and management detail, and for NSI-application guidance, see the individual `VS-0x` business-layer documents.

A note on numbering: each `VS-0x.y` you see below is both a value stream stage (*what value* is added) and a business process (*how* it is done). The numbering maps one-to-one; this document uses the business-process view. See the key-concepts box above for the distinction.

The diagram below shows the whole business layer in one picture: data enters from external providers on the left, gains value through the six primary streams, and reaches users and society on the right — with the four supporting streams enabling every step.

Steady States of Data

The streams work independently of each other (in architecture terms: they are **loosely coupled**). They exchange data only at defined **steady states** — points where a high-quality, versioned dataset is handed

[1] This split follows the ArchiMate 3.2 Specification (sec. 7.3.2): "value streams are typically realized by business processes", with the stages providing "a framework for organizing and defining business processes".

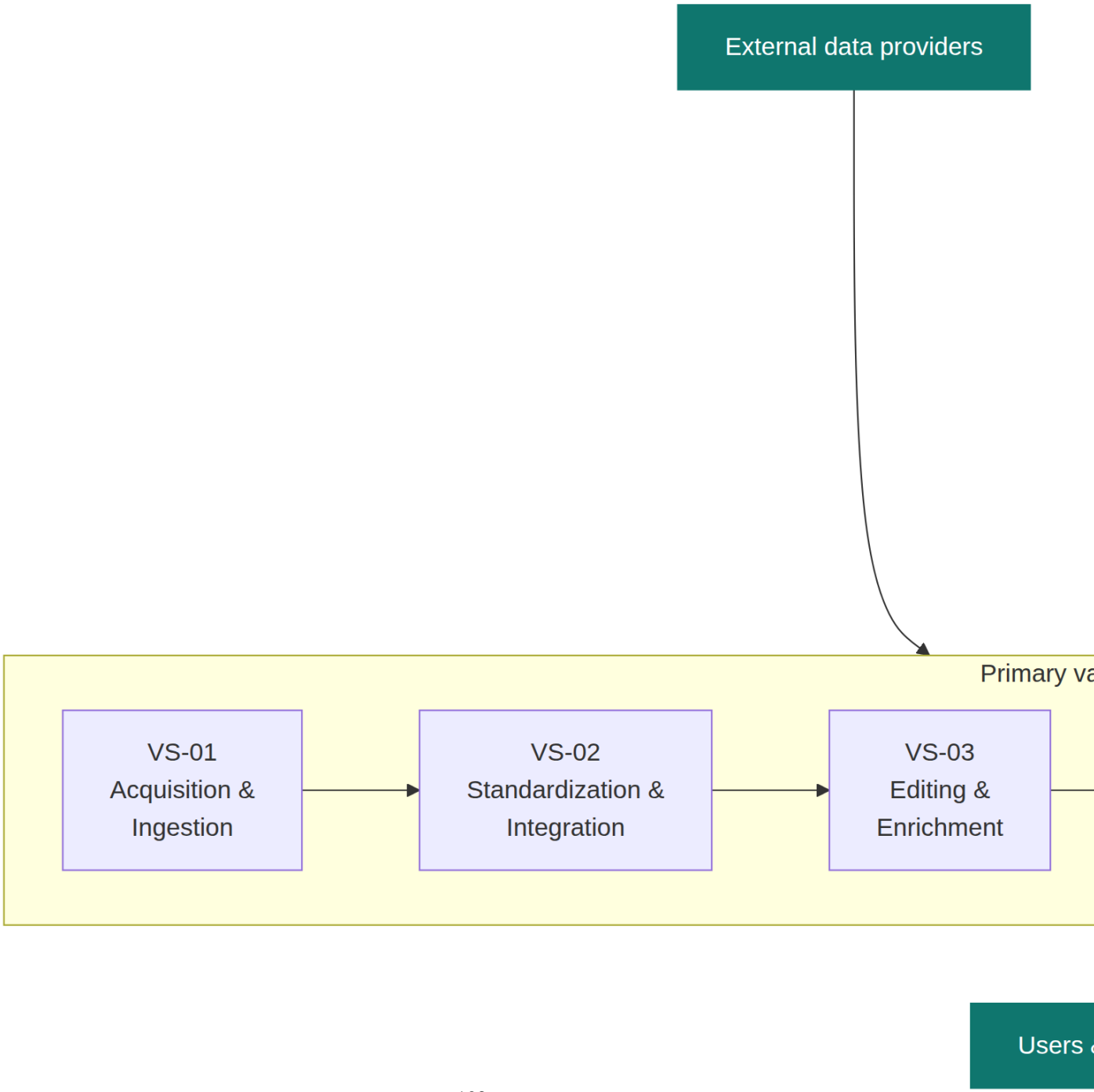


Figure 1: The six primary value streams as a production chain, delivering to users and society. Purple = the four supporting streams (they enable the chain, dashed arrow); dark green = parties outside

over. In the usual case each primary stream consumes one steady state and produces the next, so a stream needs no knowledge of its neighbors' internals (SAF-P07). The handover works through events: when a stream registers a finished dataset, a notification tells the next stream that new data is ready. Not every process has to pass through every state; see Applying the steady states pragmatically.

Of the five, **processed data (#3) is the only steady state mandatory for all production** — it is the microdata on which everything downstream is verifiably built. **Raw Data (#1)** is also mandatory when a product collects its own data, but not for a macro statistic that starts entirely from another domain's data. **Standardized data (#2)** is always optional, and **Statistics (#4)** may share its artefact with **Released data (#5)** when the two are identical.

The table lists the five numbered steady states (#1–#5) in the order data passes through them, with the two boundary positions (#0 and #6) outside the NSI's control shown for context:

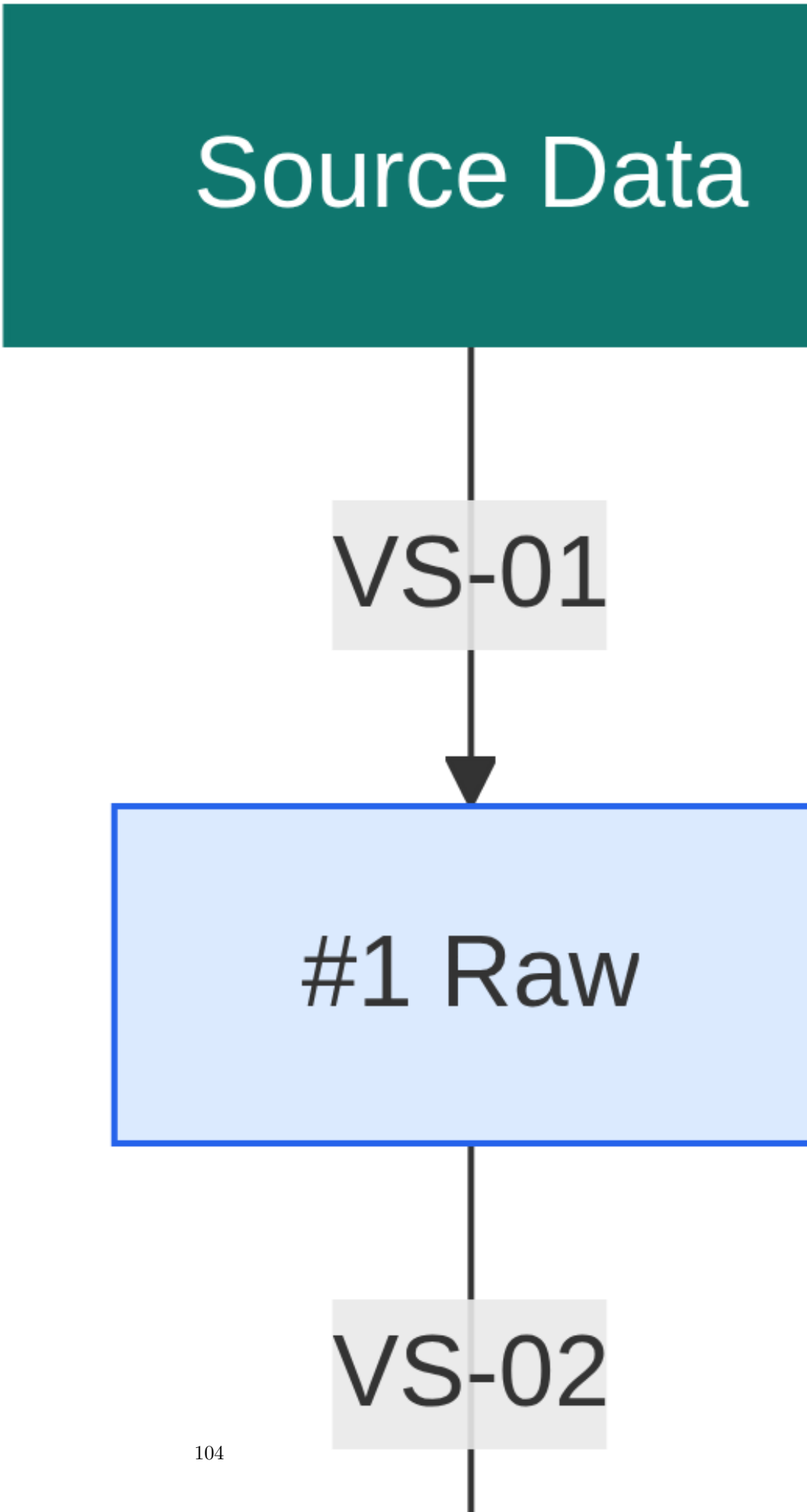
Table 1: Steady States of data

State	Name	Produced by	Meaning
(#0)	Source Data	*external providers*	Data at the external holder, before it enters the NSI (pre-ingestion; not a numbered SAF state)
#1	Raw Data	VS-01	Received into NSI custody in original form, with provenance (a record of where it came from and how) and basic validation
#2	Standardized Data	VS-02	Aligned to common structure, format, and classifications — **values unchanged**
#3	Processed Data	VS-03	Content validated, edited, imputed, enriched, integrated; derived variables added
#4	Statistics (Output Data)	VS-04 → VS-05	Estimated/aggregated statistics (VS-04), composed into approved products (VS-05)
#5	Released Data	VS-06	Disclosure-controlled, documented, published/delivered to external parties
#6	Consumed Data	*external users*	Data in use outside the NSI's direct control

Applying the steady states pragmatically

A steady state is an agreement about data: the quality it has reached, who owns it, how it is identified and what metadata comes with it. It does not require a separate physical copy at every state, and it does not require every process to pass through all six. An office may depart from the usual sequence in the three cases below, provided it records the choice (which state was shared or skipped, and why) in the catalogue entry. In other words, the rule is "comply or explain".

1. **Two states can share one artefact.** If nothing happens between State #4 and State #5 (no transformation and no further disclosure control), the #4 artefact does not have to be registered again as a new dataset; it is marked as released instead. The release steps still take place: release authorisation is obtained and the final disclosure check is carried out, and both are recorded even when the check changes nothing.
2. **A process can start downstream.** Not every production process begins at VS-01. Macro statistics, for example, are often built on Processed Data (#3) or Statistics (#4) that another team has already registered, and State #4 above can already be "shared internally for use as input data in other statistical processes". Such a process starts from the registered artefact and its lineage points to it; there is no need to create placeholder #1 or #2 artefacts. Data that comes from outside the office still enters through VS-01 and State #1.
3. **Lineage is about traceability, not duplication.** SAF-P09 requires that every value can be traced back to where it came from. When a stream uses an artefact without changing any values, a logged process event that refers to the existing artefact version meets this requirement, and no identical copy needs to be registered. As soon as a value changes, a new version is registered.


Source Data
VS-01
#1 Raw
VS-02

One limit applies to all three cases. States must not be merged where this would hide a change of values (for example, standardisation at #2 and editing at #3 carried out as a single undocumented step) or a disclosure-control step (for example, #4 and #5 when additional disclosure control is applied before release).

The Four-Process Governance Pattern

Every stream is described through the same **four process types**. This is not a one-off sequence but a repeating cycle that keeps quality improving, and each of the four produces a distinct kind of result:

- **Design** — decide how the work should be done: rules, methodology, specifications, policies.
- **Implementation** — build, configure, and test the systems that will do it.
- **Execution** — run the work each production cycle; this is what produces the statistical data.
- **Management** — check quality, monitor performance, and feed improvements back to Design.

The cycle looks like this — note the feedback arrow that closes the loop:

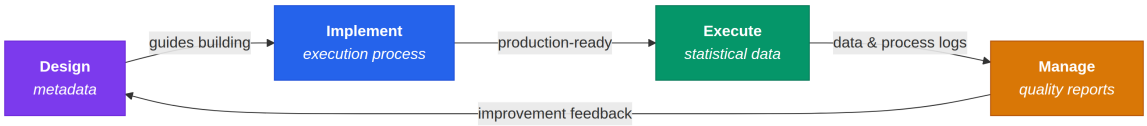


Figure 3: The four-process governance pattern. Each of the six streams runs this full cycle; the per-stream sections below summarize Design/Implementation/Management and keep the Execution detail.

> **Note**
>
> In the sections that follow, GSBPM phases are mapped per process type:
>
> - **Design → Phase 2**,
> - **Implementation → Phase 3**,
> - **Execution → the later GSBPM phases** appropriate to the domain (Phase 4 for collection, Phase 5 for processing, Phase 6 for analysis, Phase 7 for dissemination),
> - **Management → overarching Quality Management / Phase 8**.
>
> The four **supporting** streams (VS-07–VS-10) do not follow this production cadence: they run a **continuous plan → build → operate → improve** cycle and align to **GAMSO** rather than GSBPM (see the supporting-streams section).

Principles to Apply

The SAF defines eleven architecture principles, labeled SAF-P01 to SAF-P11. The six **primary** streams actively apply six of them (marked below); the four **supporting** streams foreground the other five (marked †). The table shows each principle with practical guidance on how to apply it; VS-07 in particular **enforces** the data principles (P09/P10/P11) across the whole chain.

Table 2: SAF Principles

Principle	Statement	How to apply across the value chain
SAF-P01	Actively participate in open source to maintain collective control over statistical processes	Build statistical capabilities on shared, open tooling; contribute back, so NSIs keep collective control of critical methods rather than depending on vendors.
SAF-P07	Implement loosely coupled processes and systems	Exchange data only at steady-state interfaces; each stream consumes one state and produces the next without knowing the others' internals.

Principle	Statement	How to apply across the value chain
SAF-P08	Data is shared property	Once an artefact is registered at a steady state, make it available to all authorized consumers; avoid duplicating collection, processing, or analysis.
SAF-P09	No data without metadata and classification	Every artefact carries provenance (origin), methodology, accuracy, classification, and lineage (the trace back to its inputs) — captured at the point each value is created or changed.
SAF-P10	Security By Design	Embed access control, integrity verification, and tamper-evident audit trails in every pipeline from the start.
SAF-P11	Privacy By Design	Remove/pseudonymize identifiers during standardization; assess disclosure risk on enrichment and apply statistical disclosure control before release.
SAF-P02 †	Active support	Operate and support capabilities as living services, not one-off builds (VS-09 platforms, VS-10 services).
SAF-P03 †	Active Life Cycle management	Version and govern rules, methods, models, and products across their life cycle; retire what is obsolete (VS-07, VS-08).
SAF-P04 †	Maximize the benefits for the organization	Prefer solutions that serve the whole organization and real user value over local optimizations (VS-10, VS-07).
SAF-P05 †	IT and technology is everyone's business	Treat methodology, data, and technology as a shared responsibility across business and IT (VS-08, VS-09).
SAF-P06 †	Business Continuity	Keep the chain running reliably and able to recover from disruption (VS-09).

= actively applied by the six primary value streams. † = foregrounded by the four supporting value streams (VS-07–VS-10).

! Two cross-chain disciplines

Metadata everywhere (SAF-P09): value is only added once its context is captured — what was done, by which rule/method, with what accuracy, and traceable to its inputs.
Confidentiality progressively applied (SAF-P11): identifiers are removed/pseudonymized at standardization (#2) at the latest, or earlier where that suits the process; disclosure risk is assessed on enrichment (#3), disclosure control is applied to outputs (#4), and a final disclosure check gates release (#5).

VS-01.0 — Data Acquisition and Ingestion

Domain: Observation/Collection · **State transition:** Source Data → **#1 Raw Data** · **Value:** all data enters NSI custody, verifiably and with provenance.

VS-01 is the front door of the statistical system. Its job is to bring data from the outside world — survey respondents, administrative registers, and emerging sources such as web or sensor data — into the NSI safely and with a complete record of where it came from. Nothing about the data's content is changed here; the point is to receive it, prove it is what it claims to be, and register it so the rest of the chain can find and trust it. The output is **Raw Data**: the data in its original form, under NSI custody, accompanied by provenance.

Business processes

- VS-01.1 Identify Data Needs and Sources
- VS-01.2 Establish Data Agreement and Access Right
- VS-01.3 Collect, Receive, or Extract Data
- VS-01.4 Validate and Log Data Intake
- VS-01.5 Store and Register Raw Data Artefacts

How the business processes work. The cycle starts by **identifying the data needs and sources** for the round — confirming what variables and populations are required and that the chosen sources are available and accessible. Once the source is settled, the team **establishes the data agreement and access rights**: activating or renewing the legal basis (a data-sharing agreement, or the relevant GDPR ground) and checking that credentials are valid, so collection is lawful and controlled. Data is then **collected, received, or extracted** through the appropriate channel — a survey instrument, an administrative file transfer, or an API/scraping job — and at the moment of receipt the ingestion metadata (source, method, timestamp, file hash, volume) is captured. Each delivery is **validated and logged**: completeness, structure, and integrity are checked and every result written to an audit trail, so the NSI can be confident the intake is whole and traceable. Finally the dataset is **stored and registered as a raw-data artefact** — given a persistent identifier, kept in its original form, and cataloged with its provenance — and a "Raw Data Registered" event tells VS-02 that new data is ready.

> **i** Signature discipline — no data without metadata
>
> Every dataset is enriched at ingestion with provenance, source, method, timestamp, and a quality and BIV classification — availability, integrity, confidentiality — per SAF-P09, with security and privacy applied by design (SAF-P10/P11).

Around this per-cycle run, **Design** prepares the data requirements, collection methodology, raw-data schema, validation rules, and access/privacy framework; **Implementation** builds the collection instruments and ingestion/validation workflows and tests them; and **Management** monitors response, timeliness, and rejection rates, assesses intake quality, and feeds improvements back to design.

Business process	What it produces
VS-01.1 Identify Data Needs and Sources	Confirmed requirements and available sources for the cycle
VS-01.2 Establish Data Agreement and Access Right	Active, lawful access (agreements, GDPR basis, credentials)
VS-01.3 Collect, Receive, or Extract Data	Data in NSI custody + captured ingestion metadata
VS-01.4 Validate and Log Data Intake	Validated intake + audit trail
VS-01.5 Store and Register Raw Data Artefacts	Registered Raw Data artefact + "Raw Data Registered" event

Boundaries: in — Source Data from external providers; out — **Raw Data** on *"Raw Data Registered"* → VS-02.

VS-02.0 — Data Standardization and Integration

Domain: Standardization without loss of content · **State transition:** #1 Raw → **#2 Standardized** · **Value:** coherence and comparability across sources.

VS-02 makes data from different sources speak the same language. After raw data is in custody, this stream aligns its structure, format, and codes to the NSI's agreed standards so that datasets which looked different at the source can be compared and combined. Crucially it does this **without changing any value** — it re-expresses and annotates, it does not transform. The output is **Standardized Data**: the same information, now in a common shape, expressed in standard classifications, and carrying rich metadata.

Business processes

- VS-02.1 Profile and Assess Data Structure
- VS-02.2 Apply Standard Classifications and Formats
- VS-02.3 Align Metadata and Semantics
- VS-02.4 Validate Structural and Semantic Conformance
- VS-02.5 Register Standardized Artefacts

How the business processes work. The stream first **profiles and assesses the incoming structure**, comparing it against the target data model to see which fields map directly, which simply need reformatting, which source codes have a standard correspondence, and which values are uncoded. It then **applies standard classifications and formats**: source codes are mapped one-to-one to standards (for example a source activity code to NACE, the European industry classification) through maintained correspondence tables, and the layout is coerced into the common format (renaming, restructuring, type-casting, e.g. CSV→XML). The original values are preserved, and anything that has no deterministic correspondence — or is free text needing interpretation — is flagged and passed downstream to VS-03. Next it **aligns metadata and semantics**: variables are linked to canonical concepts, and each variable's unit, currency, and reference-period basis (and any differing cross-source definitions) are *recorded as metadata* rather than converted. The result is **checked for structural and semantic conformance**, including a no-loss-of-content check confirming that record counts and key measures are unchanged. Conformant data is then **registered as a standardized artefact** with its mapping lineage, and a "Standardized Data Registered" event notifies VS-03.

> **i** Signature discipline — standardization without loss of content
>
> VS-02 only maps codes 1-to-1, reformats structure, and annotates metadata. It **never changes a value**; unit/currency conversion, reference-period alignment, value reconciliation, and interpretive free-text coding are deliberately deferred to VS-03.

Around this, **Design** defines the target data model, the deterministic classification correspondences, the harmonization and conformance rules, and the privacy framework; **Implementation** builds the mapping & format-coercion component, the classification-mapping services, the conformance pipeline, and identifier pseudonymization; **Management** tracks mapping coverage and the unmapped-value rate, governs classification versions, and improves the correspondences.

Business process	What it produces
VS-02.1 Profile and Assess Data Structure	A structure-gap assessment against the target model
VS-02.2 Apply Standard Classifications and Formats	Standard-coded, common-format data (values unchanged); unmapped values flagged for VS-03
VS-02.3 Align Metadata and Semantics	Canonical-concept links + unit/period/definition metadata
VS-02.4 Validate Structural and Semantic Conformance	Conformance result + no-loss-of-content confirmation
VS-02.5 Register Standardized Artefacts	Registered Standardized Data + "Standardized Data Registered" event

Boundaries: in — Raw Data on *"Raw Data Registered"*; out — **Standardized Data** on *"Standardized Data Registered"* → VS-03.

VS-03.0 — Data Editing and Enrichment

Domain: Editing, derivation, and non-response correction · **State transition:** #2 Standardized → **#3 Processed** · **Value:** data reliability and analytical readiness.

VS-03 is where data quality is actively improved. Standardized data is rarely clean — it contains errors, inconsistencies, gaps, and not-yet-coded values. This stream detects those problems, corrects them, fills the gaps, and derives the extra variables analysis needs. Uniquely in the chain, it **deliberately changes values to do so**. What makes that safe is total traceability: every change is recorded against its original value and a reproducible, versioned dataset is produced. The output is **Processed Data**: reliable, complete, analytically ready microdata.

Business processes

- VS-03.1 Conduct Error Detection / Quality Diagnostics
- VS-03.2 Perform Data Editing and Correction
- VS-03.3 Impute Missing or Non-Response Data

- VS-03.4 Derive Additional Variables and Indicators
- VS-03.5 Perform Quality Evaluation and Versioning

How the business processes work. The stream begins by **detecting errors and running quality diagnostics** — consistency, range, outlier, and duplicate checks — and uses selective-editing scores to prioritize the problems that actually matter for the results. It then **performs editing and correction**: low-impact issues are auto-corrected by rule while high-impact records are reviewed by subject-matter analysts, and this is where the interpretive work VS-02 deferred is carried out (coding free-text values such as occupation to ISCO, the international occupation classification, and reconciling values where integrated sources disagree). Every change logs the original value, the rule or decision, the reason, the actor, and the time. Genuine gaps are then closed by **imputing missing or non-response data** using approved methods, with each imputed value clearly flagged so it stays distinguishable from observed data. The dataset is **enriched by deriving additional variables and indicators** — and this is also where unit/currency conversion and reference-period alignment are computed from the bases VS-02 recorded, and where datasets are integrated on common keys. Finally the stream **evaluates quality and versions the result**: it measures the improvement against the incoming baseline, verifies the edit lineage is complete and reversible, and creates an immutable Processed Data version, signaled to VS-04 by a "Processed Data Registered" event.

> **i** Signature discipline — full traceability
>
> VS-03 is the domain that changes values, under the guarantee of *no undocumented or irreversible change*: original values are preserved beside edited ones, imputations are flagged, and every version is reproducible.

Around this, **Design** sets the error-detection and editing rules (with a selective-editing strategy), the imputation methodology, the derivation and coding specs, the integration rules, and the versioning policy; **Implementation** builds the editing, imputation, coding, derivation, and integration engines and the version store; **Management** watches edit, imputation, and over-editing rates, assesses quality improvement, and audits the edit lineage.

Business process	What it produces
VS-03.1 Conduct Error Detection / Quality Diagnostics	A prioritized diagnosis of errors and gaps
VS-03.2 Perform Data Editing and Correction	Corrected values + interpretive coding + full edit lineage
VS-03.3 Impute Missing or Non-Response Data	Completed data with imputation flags
VS-03.4 Derive Additional Variables and Indicators	Derived variables + unit/currency/period conversions + integrated data
VS-03.5 Perform Quality Evaluation and Versioning	Versioned Processed Data + "Processed Data Registered" event

Boundaries: in — Standardized Data on *"Standardized Data Registered"*; out — **Processed Data** on *"Processed Data Registered"* → VS-04.

VS-04.0 — Estimation, Analysis, and Statistical Modeling

Domain: Estimation and closed-loop · **State transition:** #3 Processed → **#4 Statistics** · **Value:** analytical insight and statistical credibility.

VS-04 turns clean microdata into statistics. Up to now the chain has improved the data itself; here the focus shifts to producing numbers that describe a population — estimates, weighted aggregates, indices, and model results. The microdata is not edited further; instead statistics are computed *from* it, each accompanied by a measure of how accurate it is. The output is **Output Data / Statistics**: validated figures with their methodology and accuracy attached, and an initial layer of confidentiality protection applied.

Business processes

- VS-04.1 Confirm Estimation and Analytical Approach
- VS-04.2 Apply Weighting and Calibration
- VS-04.3 Estimate and Aggregate Data
- VS-04.4 Perform Model-Based Analysis
- VS-04.5 Validate and Document Results

How the business processes work. Each round starts by **confirming the estimation and analytical approach** — selecting the pre-approved method for this cycle's data and checking that the calibration totals and domains are current. The stream then **applies weighting and calibration** to correct for sampling, non-response, and coverage, so the sample represents the population. With weights in place it **estimates and aggregates the data**, producing the target statistics together with their accuracy (standard errors and confidence intervals) — and because VS-03 flagged which values were imputed, that added uncertainty is reflected in the variance. Where direct figures are not enough, it **performs model-based analysis** (regression, time-series/forecasting, small-area models) to reveal relationships and produce reliable estimates for thin domains, recording the model diagnostics. Finally it **validates and documents the results**: checking statistical soundness and coherence with related and prior statistics, applying an initial layer of statistical disclosure control so the figures can potentially be released, documenting the methodology, and versioning the output — announced to VS-05 by an "Output Data Registered" event.

> **i** Signature discipline — methodological transparency
>
> Every estimate carries the method, parameters, and accuracy measure that produced it and can be regenerated; imputation uncertainty from VS-03 is reflected in the variance, and initial statistical disclosure control is applied so output can potentially be released.

Around this, **Design** fixes the estimation strategy and accuracy approach, the weighting/calibration methodology, the aggregation and model specifications, and the validation and disclosure-control policy; **Implementation** builds the weighting, estimation, variance, and modeling engines and the disclosure-control and version tooling; **Management** tracks accuracy, timeliness, and revisions, assesses coherence, and audits reproducibility.

Business process	What it produces
VS-04.1 Confirm Estimation and Analytical Approach	The pre-approved method confirmed for the cycle
VS-04.2 Apply Weighting and Calibration	Weights that make the sample represent the population
VS-04.3 Estimate and Aggregate Data	Estimates and aggregates, each with an accuracy measure
VS-04.4 Perform Model-Based Analysis	Model/forecast/small-area results with diagnostics
VS-04.5 Validate and Document Results	Validated, disclosure-controlled, versioned statistics + "Output Data Registered" event

Boundaries: in — Processed Data on *"Processed Data Registered"*; out — **Output Data** on *"Output Data Registered"* → VS-05.

VS-05.0 — Statistical Product Composition

Domain: Statistically composed and composite products · **State transition:** within **#4 Statistics** (analytical outputs → approved products) · **Value:** coherence and interpretability.

VS-05 assembles finished products from the statistics VS-04 produced. Individual estimates are not yet something a user reads; this stream brings results from different domains, surveys, and time periods together into coherent publications, time series, indices, and datasets — and approves them for release. It does not re-estimate anything; it selects, combines, checks, explains, and signs off. The output is an **approved, registered statistical product** ready for dissemination.

Business processes

- VS-05.1 Select and Integrate Source Datasets
- VS-05.2 Construct Composite Indicators or Indices
- VS-05.3 Perform Consistency and Coherence Checks
- VS-05.4 Generate Analytical and Interpretive Outputs
- VS-05.5 Approve and Register Statistical Products

How the business processes work. The stream first **selects and integrates the source datasets** — picking the exact validated output versions a product needs and combining them, carrying their accuracy metadata along. It then **constructs the composite indicators or indices** the product presents (ratios, indices, chained time series), deriving each composite's uncertainty from its components. The assembled product is put through **consistency and coherence checks**: totals must reconcile to their parts, and the figures must agree with related statistics and with prior releases — discrepancies are explained or sent back, never quietly adjusted. Next the stream **generates the analytical and interpretive outputs** — the commentary, key messages, tables, and visualizations that make the product understandable — and runs them through editorial review. Finally it **approves and registers the product**: validating it against dissemination and confidentiality standards, obtaining sign-off from the designated authority, and registering an immutable version with full lineage, which a "Statistical Product Registered" event passes to VS-06.

> **i** Signature discipline — coherence and interpretability
>
> VS-05 **composes** validated outputs (it does not re-estimate). Products are internally consistent, coherent with related and prior statistics, carry their components' accuracy metadata, and are **approved against dissemination and confidentiality standards** before being registered.

Around this, **Design** defines the product portfolio, the composite-indicator specs, the coherence checks, the content templates, and the approval/confidentiality/versioning framework; **Implementation** builds the integration and composition engines, the indicator/time-series engines, the coherence and content tooling, and the approval/version store; **Management** tracks coherence-check pass rates and approval iterations, assesses product quality, and audits reproducibility and approval.

Business process	What it produces
VS-05.1 Select and Integrate Source Datasets	Integrated components with accuracy carried forward
VS-05.2 Construct Composite Indicators or Indices	Composite measures and time series
VS-05.3 Perform Consistency and Coherence Checks	Confirmed internal consistency and coherence
VS-05.4 Generate Analytical and Interpretive Outputs	Reviewed narrative, tables, and visualizations
VS-05.5 Approve and Register Statistical Products	Approved, versioned product + "Statistical Product Registered" event

Boundaries: in — Output Data on *"Output Data Registered"*; out — **Approved Statistical Product** on *"Statistical Product Registered"* → VS-06.

VS-06.0 — Dissemination and Service Delivery

Domain: Dissemination (final stage) · **State transition:** #4 Approved products → **#5 Released** → #6 Consumed · **Value:** impact and accessibility.

VS-06 is the chain's exit point — where approved products become public value. It publishes and delivers statistics to policymakers, researchers, journalists, businesses, and the public through trusted, secure channels, and supports users in interpreting them correctly. The decisive control here is confidentiality: a final disclosure-control gate stands between an approved product and release. The output is **Released Data** (State #5); once in users' hands it becomes **Consumed Data** (State #6), beyond the NSI's direct control.

Business processes

- VS-06.1 Plan Dissemination and Access Strategy
- VS-06.2 Prepare Data for Dissemination
- VS-06.3 Publish and Provide Access Services
- VS-06.4 Support and Engage Users
- VS-06.5 Monitor Usage and Feedback

How the business processes work. Each release begins by **confirming the dissemination and access plan** — the channels (publication, API/open data, dashboards, secure research access), the formats, the release-calendar slot, and the embargo/equal-access arrangements. The product is then **prepared for dissemination**: final statistical disclosure control is applied (suppression, rounding, perturbation; de-identification of any microdata) and the release must clear the disclosure-risk criteria — *nothing proceeds if it does not* — after which it is formatted for each channel and packaged with its documentation and metadata. The stream then **publishes and provides access services**, releasing the product and its metadata across the confirmed channels with timing managed so that access is equal (no privileged early sight outside policy). It **supports and engages users** with documentation, a helpdesk, training, and errata, so the statistics are interpreted responsibly. Finally it **monitors usage and feedback** — tracking downloads, API calls, and dashboard views, and collecting satisfaction and questions — and routes those insights back to design, to VS-10, and across the chain. The release is signaled by a "Statistics Released" event.

> **i** Signature discipline — accessibility, trust, and the release gate
>
> Accurate, timely, understandable delivery through secure channels, gated by **final statistical disclosure control** (no release without confidentiality clearance) and supported by documentation that enables responsible, equal-access use. VS-06 relies on the supporting streams VS-07 (compliance), VS-09 (platforms), and VS-10 (user needs).

Around this, **Design** sets the dissemination and access strategy, the final disclosure-control and de-identification rules, the channels/formats/documentation standards, and the release governance; **Implementation** builds the publication, API, dashboard, and secure-research-access services (on VS-09 platforms) and the disclosure-control and release tooling; **Management** monitors punctuality, usage, and satisfaction, assesses accessibility, and audits disclosure and equal-access compliance with VS-07.

Business process	What it produces
VS-06.1 Plan Dissemination and Access Strategy	Confirmed channels, formats, and release timing
VS-06.2 Prepare Data for Dissemination	Disclosure-safe, channel-ready products + documentation
VS-06.3 Publish and Provide Access Services	Released statistics across channels, equally and on time
VS-06.4 Support and Engage Users	User documentation, helpdesk, training, errata
VS-06.5 Monitor Usage and Feedback	Usage insight + feedback routed to design and VS-10

Boundaries: in — Approved Statistical Product on *"Statistical Product Registered"*; out — **Released Data** on *"Statistics Released"* → users/society (consumed as State #6). No downstream primary stream; feedback returns to VS-10.

Supporting Value Streams (VS-07 to VS-10)

The four supporting streams are not stages in the production line — they work **across the whole chain**, enabling and constraining all six primary streams continuously. Reading their sections requires a small shift of mindset from the six above.

❗ How a supporting value stream differs from a production stream

- **No steady-state data transition.** A supporting stream does not take data from one state to the next; it does not appear in the #1→#6 chain. There is no "Data Registered" handover event and no "Boundaries: in→out" — instead each interacts with the whole chain at once.
- **It runs continuously**, as a plan → build → operate → improve cycle, not per production cycle.
- **Its output is an enabling capability, not data:** governance & assurance (07), validated methods (08), platforms (09), understood needs & relationships (10).
- **It aligns to GAMSO** — the standard model of management and support activities (Strategy & Leadership, Capability Management, Corporate Support) — rather than to GSBPM production phases, and it foregrounds principles **P02–P06**.

The diagram shows the two directions of that interaction: the supporting streams provide capabilities to the chain (arrows in), and the chain sends back evidence, needs, and usage (arrow out):

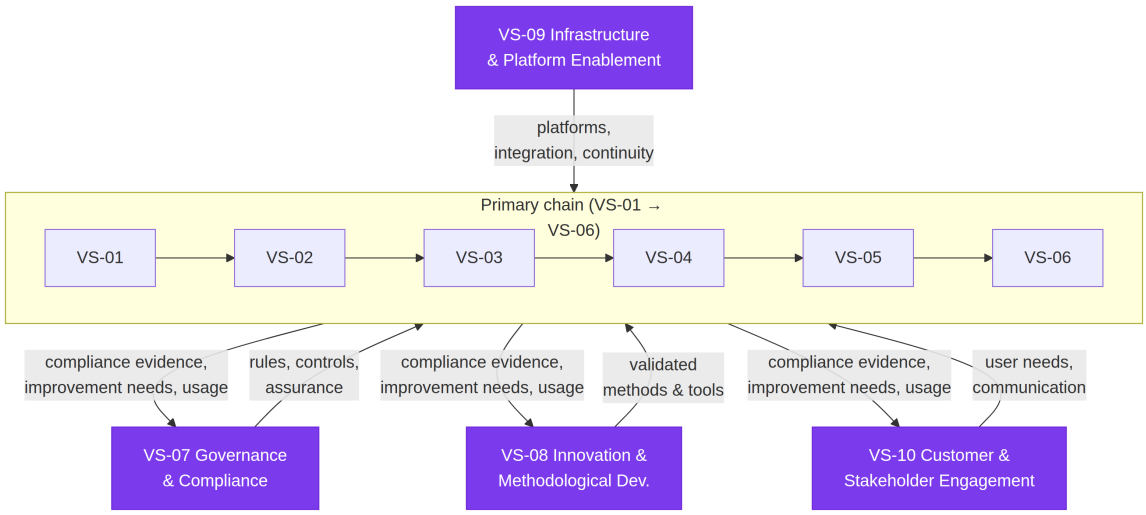


Figure 4: The four supporting streams (purple) and the primary chain: each provides a capability to the chain and receives feedback or needs from it.

VS-07.0 — Governance and Compliance

Domain: Governance (horizontal) · **GAMSO:** Strategy & Leadership + Corporate Support · **Value:** integrity and accountability.

VS-07 makes sure every process, dataset, and product operates under clear rules, standards, and accountability. It builds trust by ensuring all data activities are ethical, transparent, and compliant with legal, methodological, and quality requirements (Code of Practice, GDPR, ISO). It does not move data along the chain — it **oversees** the streams that do, and is the place where the data principles (P09/P10/P11) are enforced organization-wide.

Business processes

- VS-07.1 Define Governance Frameworks and Policies
- VS-07.2 Establish Legal and Regulatory Compliance Mechanisms
- VS-07.3 Implement Data and Metadata Governance
- VS-07.4 Monitor Quality and Compliance Indicators
- VS-07.5 Audit, Certify, and Report Compliance
- VS-07.6 Update and Improve Governance Mechanisms

How the business processes work. Governance runs as a continuous loop. It first **defines governance frameworks and policies** (07.1) — the charters and principles covering data, metadata, ethics, confidentiality, and quality — and **establishes legal and regulatory compliance mechanisms** (07.2), mapping

applicable regulation to concrete procedures. These are made operational by **implementing data and metadata governance** (07.3): assigning the owner/steward/custodian roles and the metadata and lineage processes that wire governance into how the streams work. It then **monitors quality and compliance indicators** (07.4) on dashboards and periodically **audits, certifies, and reports compliance** (07.5), producing the assurance that the statistics can be trusted. Finally it **updates and improves the governance mechanisms** (07.6), folding regulatory change and lessons learned back into the policies. *(The strategy model labels two steps `VS-07.5`; the second is treated here as VS-07.6.)*

Interaction with the chain: provides frameworks, roles, and controls to all streams and receives their compliance evidence, quality indicators, and audit findings (each primary stream's audit-and-compliance reporting feeds here); issues certification and assurance to regulators and the public.

Business process	What it produces
VS-07.1 Define Governance Frameworks and Policies	Governance charters and policies
VS-07.2 Establish Legal and Regulatory Compliance Mechanisms	Compliance procedures mapped to regulation
VS-07.3 Implement Data and Metadata Governance	Assigned roles + metadata/lineage processes
VS-07.4 Monitor Quality and Compliance Indicators	Quality & compliance dashboards
VS-07.5 Audit, Certify, and Report Compliance	Audit reports, certifications, assurance
VS-07.6 Update and Improve Governance Mechanisms	Revised policies and controls

VS-08.0 — Innovation and Methodological Development

Domain: Innovation (horizontal) · **GAMSO:** Capability Management · **Value:** continuous improvement / future-proofing.

VS-08 keeps the organization from standing still: it discovers, tests, and introduces new data sources, methods, tools, and technologies, connecting research, methodology, IT, and business users. It keeps innovation **controlled** so that only sound, documented, compliant methods reach production. It does not transform data — it improves the methods and tools the production streams use.

Business processes

- VS-08.1 Identify Innovation Needs and Opportunities
- VS-08.2 Conduct Research and Proof-of-Concepts
- VS-08.3 Evaluate Methodological Soundness and Feasibility
- VS-08.4 Develop / Document New Methodologies or Tools
- VS-08.5 Integrate Innovation into Production Environments
- VS-08.6 Promote Knowledge Sharing and Capability Building

How the business processes work. The pipeline begins by **identifying innovation needs and opportunities** (08.1) — scanning technology and methodology (AI, ML, new sources) and the chain's own improvement backlog — then runs **research and proofs-of-concept** (08.2) in limited scope. Promising results pass through **evaluation of methodological soundness and feasibility** (08.3), the gate that stops attractive-but-unsound ideas, and are then **developed and documented into methods or tools** (08.4) — standardized procedures and reference implementations ready to hand over. The defining step is **integrating innovation into production** (08.5): the governed transfer of a validated method into the relevant primary stream (after VS-07 approval, on VS-09 platforms). Finally it **promotes knowledge sharing and capability building** (08.6), disseminating guidance and growing staff competency — which surfaces the next round of needs.

Interaction with the chain: receives improvement needs from every primary stream (their continuous-improvement loops) and user-driven needs from VS-10; **delivers validated methods and tools into production** through the 08.5 transfer gate.

Business process	What it produces
VS-08.1 Identify Innovation Needs and Opportunities	Prioritized innovation backlog
VS-08.2 Conduct Research and Proof-of-Concepts	Experiment / pilot results
VS-08.3 Evaluate Methodological Soundness and Feasibility	Soundness & feasibility assessment
VS-08.4 Develop / Document New Methodologies or Tools	Standardized method + reference implementation
VS-08.5 Integrate Innovation into Production Environments	Method/tool live in a production stream (governed transfer)
VS-08.6 Promote Knowledge Sharing and Capability Building	Disseminated guidance + trained staff

VS-09.0 — Infrastructure and Platform Enablement

Domain: Infrastructure (horizontal) · **GAMSO:** Capability Management + Corporate Support · **Value:** reliable enablement at scale.

VS-09 provides the technical and digital foundation every other stream runs on — environments, data platforms, tools, and integration services — keeping them secure, resilient, and compliant while flexible enough to adopt new technology. It is the business-layer view of providing the application and technology enablement the primary documents scoped out.

Business processes

- VS-09.1 Define Digital and Technical Requirements
- VS-09.2 Design Platform and Infrastructure Architecture
- VS-09.3 Provision and Configure Platforms
- VS-09.4 Enable Integration and Orchestration Services
- VS-09.5 Ensure Performance, Security, and Continuity
- VS-09.6 Evolve and Modernize the Technical Ecosystem

How the business processes work. The lifecycle begins by **defining digital and technical requirements** (09.1) from business capabilities and data volumes, feeding the **design of the platform and infrastructure architecture** (09.2) — blueprints for storage, processing, integration, and secure access. The architecture is realized by **provisioning and configuring platforms** (09.3) with governance, monitoring, and security controls, and by **enabling integration and orchestration services** (09.4) — the APIs, workflow engines, and the event interfaces the steady-state handovers rely on. Day to day it **ensures performance, security, and continuity** (09.5): monitoring, data protection, incident management, and disaster recovery. Finally it **evolves and modernizes the technical ecosystem** (09.6), adopting new technology (often from VS-08) and retiring the obsolete.

Interaction with the chain: provides environments, platforms, integration, and assured availability to all streams (including VS-06's portals/APIs and VS-08's sandboxes); receives requirements, capacity demand, and incidents; its security and continuity posture is governed by VS-07.

Business process	What it produces
VS-09.1 Define Digital and Technical Requirements	Infrastructure requirements from business needs
VS-09.2 Design Platform and Infrastructure Architecture	Architecture blueprints
VS-09.3 Provision and Configure Platforms	Deployed, controlled environments
VS-09.4 Enable Integration and Orchestration Services	APIs, workflow/automation, event interfaces
VS-09.5 Ensure Performance, Security, and Continuity	Monitoring, security, availability, DR
VS-09.6 Evolve and Modernize the Technical Ecosystem	Modernization roadmap and adopted technology

VS-10.0 — Customer and Stakeholder Engagement

Domain: Engagement (horizontal) · **GAMSO:** Strategy & Leadership + Corporate Support · **Value:** relevance and public value.

VS-10 keeps the organization connected, relevant, and responsive — building mutual understanding and trust with users, data providers, partners, and policymakers. It is a continuous two-way dialogue: it does not move data, it moves **understanding** — needs in, communication and services out.

Business processes

- VS-10.1 Identify and Segment Stakeholders
- VS-10.2 Collect and Analyze User Needs
- VS-10.3 Define and Manage Service Portfolios
- VS-10.4 Deliver Services and Manage Relationships
- VS-10.5 Engage and Communicate with Stakeholders
- VS-10.6 Monitor Impact and Satisfaction

How the business processes work. The cycle begins by **identifying and segmenting stakeholders** (10.1) and **collecting and analyzing user needs** (10.2), translating them into data or service requirements. Those needs shape the **definition and management of service portfolios** (10.3) — catalogs of services and products aligned to strategy. Services reach users through **delivering services and managing relationships** (10.4) — support desks, account management, satisfaction monitoring — alongside **engaging and communicating with stakeholders** (10.5) via outreach and knowledge-sharing. Finally it **monitors impact and satisfaction** (10.6), measuring use and societal impact and feeding insight back into planning and innovation.

Interaction with the chain: bidirectional with VS-01 (the data providers it engages) and VS-06 (whose usage and feedback it receives); it routes user needs into design across the primary streams and into VS-08's innovation backlog.

Business process	What it produces
VS-10.1 Identify and Segment Stakeholders	Stakeholder map and segments
VS-10.2 Collect and Analyze User Needs	Needs register / service requirements
VS-10.3 Define and Manage Service Portfolios	Service & product portfolio
VS-10.4 Deliver Services and Manage Relationships	Operating support & managed relationships
VS-10.5 Engage and Communicate with Stakeholders	Outreach, communication, awareness
VS-10.6 Monitor Impact and Satisfaction	Impact & satisfaction insight (fed back to the chain)

The Value Chain at a Glance

Use these two tables as a quick reference: the first summarizes the six production streams, the second the four supporting streams.

VS	Domain	State in → out	Trigger out	Signature discipline
01	Acquisition & Ingestion	Source → #1 Raw	Raw Data Registered	Metadata & validation at ingestion
02	Standardization & Integration	#1 → #2 Standardized	Standardized Data Registered	No loss of content: map, reformat, annotate
03	Editing & Enrichment	#2 → #3 Processed	Processed Data Registered	Full traceability of every value change

VS	Domain	State in → out	Trigger out	Signature discipline
04	Estimation & Modeling	#3 → #4 Statistics	Output Data Registered	Methodological transparency + accuracy + disclosure control on outputs
05	Product Composition	within #4 (→ approved)	Statistical Product Registered	Coherence & interpretability; approval before registration
06	Dissemination & Service Delivery	#4 → #5 Released → #6	Statistics Released	Accessibility & trust; final disclosure-control gate

The four supporting streams work differently — they have no state transition; they run horizontally and continuously:

VS	Domain	GAMSO area	What it provides to the chain	Principles foregrounded
07	Governance & Compliance	Strategy & Leadership; Corporate Support	Frameworks, controls, assurance; enforces the data principles	P03, P04 (+ enforces P09/P10/P11)
08	Innovation & Methodological Development	Capability Management	Validated methods & tools transferred into production	P01, P05, P03, P04
09	Infrastructure & Platform Enablement	Capability Management; Corporate Support	Platforms, environments, integration, continuity	P10, P06, P02, P05
10	Customer & Stakeholder Engagement	Strategy & Leadership; Corporate Support	Understood needs, service portfolio, relationships, impact	P04, P08, P02

> **Note**
>
> This consolidated view omits the per-stream **"Applying This to Your NSI"** maturity checklists and scenarios. For those, and for the full Design / Implementation / Management detail, diagrams, and boundary-interface tables, see the individual documents `VS-01.0` … `VS-10.0` in this folder.